Models of Neural Systems

Lecture Notes in Computational Neuroscience

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Spring 2026

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Preface

These notes grew out of the Models of Neural Systems course at the Bernstein Center for Computational Neuroscience in Berlin. The course bridges neurobiology and mathematics, building from the simplest models of neural computation up through the biophysics of ion channels, the dynamical systems theory of excitability, and the spatial and network models that connect single neurons to circuits.

The goal of these notes is not exhaustive coverage but *deep understanding*. Each chapter is built around a small number of core ideas, developed from physical intuition through to mathematical formalism. Derivations are shown in full not because the algebra is the point, but because following the logic from assumptions to conclusions is how real understanding forms.

The presentation draws on classic references in theoretical and computational neuroscience [DA01; Ger+14; Koc99; Izh07; HH52].

Throughout, we emphasize three recurring themes:

- **What is the modeling goal?** Every equation exists to answer a question about the nervous system.
- **What are the key variables and parameters?** Identifying state variables and understanding what each parameter controls is the foundation of model intuition.
- **What qualitative behavior does the model explain?** Thresholds, tuning curves, oscillations, propagation, memory—the real payoff of mathematical modeling is explaining *why* these phenomena occur.

These notes are meant to be read actively. Work through the derivations with pen and paper. Attempt the conceptual questions at the end of each chapter before discussing them. When you encounter a key equation, ask yourself: what does each term do? What happens if I change a parameter? What breaks if I remove a term?

Berlin, February 2026

Notation

General Conventions

Symbol	Meaning
$\mathbf{x}$, $\mathbf{w}$	Column vectors (boldface lowercase)
$\mathbf{M}$, $\mathbf{W}$	Matrices (boldface uppercase)
$\dot{x}$, $\dot{V}$	Time derivative $\mathrm{d}x/\,\mathrm{d}t$
$\frac{\partial f}{\partial x}$	Partial derivative
$\mathrm{d}x$	Differential
$\mathbb{R}$	Real numbers
$\mathbb{E}[\cdot]$	Expectation

Electrophysiology

Symbol	Units	Meaning
V, V_m	mV	Membrane potential
V_{rest}	mV	Resting membrane potential
V_{th}	mV	Spike threshold voltage
V_{reset}	mV	Post-spike reset voltage
E_{X}	mV	Reversal (Nernst) potential of ion X
$\bar{g}_{\text{X}}$	S (or nS)	Maximal conductance of ion X
g_L	S (or nS)	Leak conductance
C_m	F (or pF)	Membrane capacitance
τ_m	s (or ms)	Membrane time constant C_m/g_L
τ_{ref}	ms	Refractory period
m, h, n	—	HH gating variables (activation/inactivation)
$m_\infty(V)$	—	Steady-state activation
$\tau_x(V)$	ms	Voltage-dependent gating time constant
$\alpha_x(V)$, $\beta_x(V)$	ms^{-1}	Opening/closing rate functions

Cable Theory

Symbol	Units	Meaning
a	cm	Fiber radius
r_L	$\Omega\,\mathrm{cm}$	Axial (intracellular) resistivity
r_m	$\Omega\,\mathrm{cm}^2$	Specific membrane resistance
c_m	$\mathrm{F/cm}^2$	Specific membrane capacitance
λ	cm	Electrotonic length constant

Network Models

Symbol	Units	Meaning
$\mathbf{u}$	—	Input vector
$\mathbf{v}$	—	Network activity vector
$\mathbf{W}$	—	Feedforward weight matrix
$\mathbf{M}$	—	Recurrent weight matrix
λ_μ	—	Eigenvalue of $\mathbf{M}$
$\mathbf{e}_\mu$	—	Eigenvector of $\mathbf{M}$
τ_r	ms	Network time constant

Physical Constants

Symbol	Value	Meaning
R	$8.314\,\mathrm{J\,mol^{-1}K^{-1}}$	Gas constant
F	$96{,}485\,\mathrm{C\,mol^{-1}}$	Faraday constant
k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}}$	Boltzmann constant
T	310 K (37 °C)	Body temperature
RT/F	≈ 26.7 mV (at 310 K)	Thermal voltage

Part I

Neural Computation and Learning

1

Simple Units and Networks

Before we can model neurons, we need to decide what a model is for. This chapter introduces the simplest mathematical neuron—the perceptron—and asks what computations a single unit or a small network can perform. The central ideas are classification, linear separability, and the dot product as a measure of similarity.

1.1 Why Model?

A model is a simplified mathematical description of a complex system. The point is not to reproduce every biological detail—that would just be a second copy of the brain, equally hard to understand. The point is to isolate the mechanisms that matter for a particular question and to make predictions that can be tested.

What makes a model "good" depends entirely on the question. A model of how ion channels generate an action potential needs biophysical parameters: conductances, gating kinetics, reversal potentials. A model of how the brain classifies visual objects needs none of that—it needs a description of how inputs are weighted and combined. Both are valid. The art is choosing the right level of abstraction.

Computational neuroscience operates across many scales: from the stochastic opening of a single channel, to the electrical dynamics of a single neuron, to the collective activity of thousands of neurons in a network, to the behavior of an animal. At each scale, different state variables and different equations are appropriate. This course begins at the simplest level—a single unit that takes inputs and produces an output—and works upward.

A useful way to organize these scales is by asking what is treated as a state variable and what is treated as hidden detail. At the molecular scale, state variables are channel states and ionic concentrations. At the single-cell scale, they are voltage and gating variables. At the population scale, they are firing rates or spike counts of groups of neurons. At the behavioral scale, they are latent decision variables and motor outputs. The same biological system can therefore admit multiple valid models, each with different observables, equations, and predictive targets.

Computationally, this layered view prevents a common mistake: using a model at the wrong level for the question at hand. We will repeatedly move between levels in later chapters

(from channels to spikes to networks), but always with explicit assumptions about what has been coarse-grained away.

1.2 Classification as a Computational Task

Many things the brain does can be framed as classification: given some sensory input, assign it to a category. Is this object a face or not? Is this sound a vowel or a consonant? Should I move left or right?

Abstractly, we have a feature vector $\mathbf{x} \in S_x$ and a set of labels S_y. Classification means finding a function $f : S_x \to S_y$ that maps inputs to correct labels. The challenge is that the space of possible mappings is enormous. If each input is a binary vector of length n, there are 2^n possible inputs and $|S_y|^{2^n}$ possible functions. Even for modest n, this number is far too large for a lookup table.

The brain solves this problem efficiently. There is evidence that higher cortical areas store something like prototypes—internal representations of typical examples for each category. A new input is compared to these prototypes, and the closest match determines the classification. The question is: what is the simplest mathematical device that can do this?

1.3 The Perceptron

The perceptron is the simplest model of a neuron that can perform classification. It takes n inputs, computes a weighted sum, and passes the result through a nonlinear function:

The Perceptron

$$y = f\Big(\sum_{i=1}^{n} w_i x_i - \theta\Big) \tag{1.1}$$

x_i: input (firing rate of neuron i), w_i: synaptic weight, θ: threshold, f: transfer function.

The weighted sum $h = \sum_i w_i x_i$ is called the *activation*. The transfer function f determines the output format. Common choices include the step function (McCulloch–Pitts neuron), which outputs 1 if $h > \theta$ and 0 otherwise, and smooth alternatives like the sigmoid $f(z) = 1/(1+e^{-z})$.

What does the perceptron actually compute? Writing the activation in vector notation, $h = \mathbf{w} \cdot \mathbf{x}$, we see that it is a *dot product*. The output is large when the input $\mathbf{x}$ points in roughly

the same direction as the weight vector $\mathbf{w}$, and small or zero when they point in different directions. The perceptron is, at its core, a similarity detector: it measures how closely the current input matches the pattern stored in the weights.

1.3.1 Decision boundaries

The threshold condition $\mathbf{w} \cdot \mathbf{x} = \theta$ defines a *hyperplane* in input space. On one side the perceptron outputs 1; on the other side it outputs 0. The weight vector $\mathbf{w}$ is perpendicular to this hyperplane, and the threshold θ determines how far the hyperplane sits from the origin.

For a two-class problem, the perceptron divides input space into two half-spaces. Any input falling on the $\mathbf{w}$ side of the hyperplane is classified as "yes"; anything on the other side is classified as "no." This geometric picture makes the perceptron's power and limitations immediately clear.

1.4 Linear Separability

A classification problem is *linearly separable* if there exists a hyperplane that correctly separates all inputs of one class from all inputs of the other. The perceptron can solve a problem if and only if it is linearly separable.

Many problems are linearly separable. If two categories of objects differ systematically along some combination of features, a single hyperplane can separate them. But some problems are not. The canonical example is the XOR function: given two binary inputs $x_1, x_2 \in \{0, 1\}$, output 1 if exactly one input is 1, and 0 otherwise. No single line in the (x_1, x_2) plane can separate the $(0, 0)$ and $(1, 1)$ points (both class 0) from the $(0, 1)$ and $(1, 0)$ points (both class 1). The perceptron fails here.

This is not a minor technicality. It tells us something fundamental: a single neuron with a linear decision boundary cannot represent arbitrary input–output relationships. To go further, we need either more neurons or a different architecture.

1.5 The Dot Product as Similarity

The fact that the perceptron computes a dot product deserves emphasis, because this operation appears throughout neuroscience and machine learning. If both $\mathbf{w}$ and $\mathbf{x}$ are normalized to unit length, then $\mathbf{w} \cdot \mathbf{x} = \cos \alpha$, where α is the angle between them. The dot product is maximal when the two vectors are aligned and zero when they are orthogonal.

This gives a natural interpretation of what the weights represent: the weight vector $\mathbf{w}$ is a *template* or prototype. The perceptron fires when the input is sufficiently similar to the template. This is consistent with the neuroscience idea that certain neurons act as feature detectors—cells in inferotemporal cortex, for instance, respond selectively to specific objects or faces. The weight vector of such a neuron encodes its preferred stimulus.

Winner-take-all competition among multiple perceptrons, each with a different weight vector, implements nearest-prototype classification. The input is assigned to the category whose prototype it most closely resembles. This is one of the simplest models of categorical perception.

1.6 Multilayer Networks

The limitations of a single perceptron disappear when we stack multiple layers. A *multilayer feedforward network* passes the input through one or more hidden layers before the output. Each hidden unit computes a weighted sum plus nonlinearity, and the outputs of one layer become the inputs to the next.

Why does depth help? Each hidden unit creates its own hyperplane in input space. The combination of many hyperplanes can carve out arbitrarily complex decision regions. A two-layer network with enough hidden units can separate any set of points—it is not limited to linearly separable problems.

This idea is made precise by the *universal approximation theorem*: a feedforward network with a single hidden layer of sufficient width and a non-polynomial activation function can approximate any continuous function on a compact domain to arbitrary accuracy. The key requirement is that the activation function is nonlinear. A network of purely linear units, no matter how deep, can only compute linear functions—the composition of linear maps is still linear.

The intuition is straightforward. Each hidden unit with a nonlinear activation (say, a ReLU) creates a "kink" in the function along some hyperplane. By rotating, shifting, and summing many such kinks, the network can sculpt any continuous shape. More hidden units means a finer partition of input space and a closer approximation to the target function.

Of course, the universal approximation theorem says nothing about how to *find* the right weights. That is the problem of learning, which we take up in Chapter 2.

Chapter Summary

Concept	Equation	Key Insight
Perceptron	$y = f\left(\sum_i w_i x_i - \theta\right)$	Weighted sum + nonlinearity
Dot product	$h = \mathbf{w} \cdot \mathbf{x}$	Measures similarity between input and template
Decision boundary	$\mathbf{w} \cdot \mathbf{x} = \theta$	Hyperplane in input space
Linear separability	—	Perceptron solves only linearly separable problems
Universal approximation	—	Sufficient width + nonlinearity $\Rightarrow$ arbitrary functions

Conceptual Questions

Q1.1 What makes a model "good" in computational neuroscience? Is a model that reproduces every biological detail necessarily better than a simpler one?

Q1.2 A perceptron with weight vector $\mathbf{w}$ classifies inputs by computing $\mathbf{w} \cdot \mathbf{x}$. If two inputs $\mathbf{x}_1$ and $\mathbf{x}_2$ produce the same activation h, what can you say about their geometric relationship to $\mathbf{w}$?

Q1.3 The perceptron's decision boundary is the hyperplane $\mathbf{w} \cdot \mathbf{x} = \theta$. How does changing θ (without changing $\mathbf{w}$) move this boundary? How does changing the magnitude of $\mathbf{w}$ (without changing its direction or θ) affect classification?

Q1.4 Why can't a single perceptron solve the XOR problem? Give a geometric argument using the decision boundary.

Q1.5 A classification problem has n-dimensional binary inputs. How many possible input–output mappings exist? Why does this make a lookup-table approach infeasible even for moderate n?

Q1.6 The dot product $\mathbf{w} \cdot \mathbf{x}$ can be interpreted as a similarity measure. Under what conditions is this interpretation most natural? What happens to the interpretation when $\mathbf{w}$ and $\mathbf{x}$ are not normalized?

Q1.7 Compare the McCulloch–Pitts step function with a smooth sigmoid as the transfer function f. What does each choice imply about how the neuron responds to inputs near the decision boundary?

Q1.8 If all activation functions in a multilayer network were linear (i.e., $f(z) = z$), the network could only compute linear functions regardless of its depth. Why? What does this tell you about the role of nonlinearity?

Q1.9 The universal approximation theorem guarantees that a wide enough single-hidden-layer network can approximate any continuous function. Does this mean a single hidden layer is always sufficient in practice? What might deeper networks offer that wider shallow networks do not?

Q1.10 Consider a winner-take-all network of multiple perceptrons, each with a different weight vector. Describe in your own words how this implements nearest-prototype classification. What biological structure might this correspond to?

2

Learning and Memory

A neural network is useless without a rule for setting its weights. This chapter introduces the Hebbian learning rule, analyzes the capacity of a single perceptron as an associative memory, and then asks how sparse coding can dramatically increase storage in a network with binary synapses.

2.1 Plasticity and Learning Rules

The weights in a neural network are not fixed—they change with experience. This is *synaptic plasticity*, and it is the physical basis of learning and memory. The question for modelers is: what rule governs how weights change?

Three broad paradigms exist. In *supervised learning*, a teacher provides the correct output for each input, and the weights are adjusted to reduce the error. In *unsupervised learning*, no teacher is available; the network discovers statistical structure in the inputs on its own. In *reinforcement learning*, the network receives a scalar reward signal and must figure out which actions led to reward. All three paradigms appear in neuroscience, but the oldest and most influential idea is Hebbian learning, which is unsupervised.

2.2 Hebbian Learning

In 1949, Donald Hebb proposed a remarkably simple principle: if neuron A repeatedly helps fire neuron B, the synapse from A to B should get stronger. The modern shorthand is "cells that fire together, wire together." Mathematically, the simplest Hebbian rule is

$$\Delta w_i = \eta \, x_i \, y \tag{2.1}$$

where x_i is the presynaptic activity, y is the postsynaptic activity, and η is a small learning rate. The weight increases when pre- and postsynaptic neurons are coactive and does not change when either is silent.

This rule is *local*: each synapse only needs information available at its own location—the activities of the neurons it connects. No global error signal is required. This is biologically attractive, because a synapse in the brain cannot easily access information from distant parts of the network.

2.2.1 Learning an association

Hebbian learning can link two previously unrelated stimuli. Suppose a visual neuron already responds to the sight of food (its weights from visual inputs are already strong). Now we present the sight and smell together. The visual stimulus drives the neuron to fire, and because the odor inputs are coactive with the output, their weights increase. After training, the odor alone can drive the neuron. The network has learned an association.

This is the computational analogue of classical conditioning: pairing a conditioned stimulus with an unconditioned one transfers the response. The Hebbian rule provides a synaptic mechanism.

2.3 The Perceptron as Associative Memory

We now ask a precise quantitative question: how many input–output associations can a single perceptron store using Hebbian learning?

Suppose we have N_s patterns to store. Each pattern is a pair $(\mathbf{u}^{(m)}, v^{(m)})$, where $\mathbf{u}^{(m)} \in \{-1, +1\}^{N_u}$ is a random binary input vector and $v^{(m)} \in \{-1, +1\}$ is the desired output. The Hebbian weight vector is the sum of outer products:

Hebbian Weight Vector

$$\mathbf{w} = \frac{1}{N_u} \sum_{m=1}^{N_s} v^{(m)} \, \mathbf{u}^{(m)} \tag{2.2}$$

$\mathbf{u}^{(m)}$: input pattern m, $v^{(m)}$: desired output, N_u: input dimension, N_s: number of stored patterns.

Each stored association contributes a term $v^{(m)}\mathbf{u}^{(m)}$ to the weight vector. The normalization by N_u keeps the weights from growing with input dimension.

2.3.1 Signal and noise

What happens when we present a stored input $\mathbf{u}^{(n)}$? The activation is

$$\mathbf{w} \cdot \mathbf{u}^{(n)} = v^{(n)} + \eta^{(n)} \tag{2.3}$$

where the first term is the correct output (the "signal") and the second term is crosstalk from all the other stored patterns:

$$\eta^{(n)} = \frac{1}{N_u} \sum_{m \neq n} v^{(m)} (\mathbf{u}^{(m)} \cdot \mathbf{u}^{(n)}) \tag{2.4}$$

SIGNAL–NOISE DECOMPOSITION

Expand $\mathbf{w} \cdot \mathbf{u}^{(n)}$ using Equation (2.2):

$$\begin{aligned}
\mathbf{w} \cdot \mathbf{u}^{(n)} &= \frac{1}{N_u} \sum_{m=1}^{N_s} v^{(m)} (\mathbf{u}^{(m)} \cdot \mathbf{u}^{(n)}) \\
&= \frac{1}{N_u} v^{(n)} \underbrace{(\mathbf{u}^{(n)} \cdot \mathbf{u}^{(n)})}_{=N_u} + \frac{1}{N_u} \sum_{m \neq n} v^{(m)} (\mathbf{u}^{(m)} \cdot \mathbf{u}^{(n)}) \\
&= v^{(n)} + \eta^{(n)}
\end{aligned}$$

The signal term is exactly $v^{(n)}$ because $\mathbf{u}^{(n)} \cdot \mathbf{u}^{(n)} = N_u$ (each component is ± 1). The noise term $\eta^{(n)}$ is a sum of $N_s - 1$ random dot products.

The classification is correct when $\text{sign}(\mathbf{w} \cdot \mathbf{u}^{(n)}) = v^{(n)}$, which requires $|\eta^{(n)}| < 1$. The question is: how likely is this to hold?

2.3.2 Noise statistics and capacity

Each dot product $\mathbf{u}^{(m)} \cdot \mathbf{u}^{(n)}$ is a sum of N_u independent random ± 1 terms. When patterns are random and uncorrelated, the central limit theorem tells us that $\eta^{(n)}$ is approximately Gaussian with mean zero and variance

$$\sigma_\eta^2 = \frac{N_s - 1}{N_u} \tag{2.5}$$

The noise grows with the number of stored patterns and shrinks with the input dimension. Retrieval is reliable when $\sigma_\eta \ll 1$, which requires $N_s \ll N_u$. A common rule of thumb is that performance remains good up to roughly $N_s \approx 0.2\, N_u$.

This is the *capacity* of the Hebbian perceptron. It is far from optimal—the Hebbian rule wastes capacity because it treats all patterns equally and does not try to minimize interference. With a more sophisticated learning algorithm (for instance, the perceptron learning rule with error correction), the capacity can be pushed up to $N_s \approx N_u$ for large N_u. But the Hebbian rule has the virtue of being local, one-shot, and biologically plausible.

2.4 Sparse Coding and Binary Synapses

We now consider a different regime: a network with binary synapses ($w_{ij} \in \{0, 1\}$) and sparse activity patterns. This model, closely related to the Willshaw network, can store far more patterns per synapse than the dense Hebbian scheme.

2.4.1 Setup

The network has N neurons in both input and output populations, with random connectivity: each pair of neurons has a synapse with probability ρ. Each pattern activates M out of N neurons, with $M \ll N$ (sparse coding). The learning rule is simple: set $w_{ij} = 1$ if both neuron i (input) and neuron j (output) are active during the presentation of a pattern. A synapse, once turned on, is never turned off.

Each association imprints at most M^2 synapses. The total number of synapses available is ρN^2. A crude upper bound on the number of storable patterns is therefore

$$P \leq \rho \left(\frac{N}{M}\right)^2 \tag{2.6}$$

2.4.2 The error criterion

The real constraint comes from false positives: a neuron that should be silent but fires because too many of its synapses have been turned on by other patterns. The probability that a particular potential synapse has been set to 1 by at least one of the P stored patterns grows with P. For an output neuron that should be off, we need all M of its incoming active synapses to happen to be zero—otherwise it could falsely activate.

The expected number of erroneously active output neurons scales as Np_{on}^M, where p_{on} denotes the probability that a given potential synapse has been switched on. Requiring fewer than one error on average gives the criterion

$$Np_{\text{on}}^M < 1 \tag{2.7}$$

Taking logarithms and solving for M:

$$M > \frac{-\ln N}{\ln p_{\text{on}}} \tag{2.8}$$

Since $\ln p_{\text{on}} < 0$ (because $p_{\text{on}} < 1$), the right-hand side is positive. Sparser patterns (smaller M) increase the capacity in Equation (2.6), but M cannot be too small or Equation (2.7) is violated. The optimal sparsity balances these two forces.

2.4.3 The sparsity–capacity tradeoff

Substituting the optimal M back into the capacity formula gives a maximum storage that scales as

$$P_{\max} \propto \frac{\rho(\ln p_{\text{on}})^2 N^2}{(\ln N)^2} \tag{2.9}$$

This is a remarkable result. The capacity grows as N^2 (up to logarithmic corrections), far faster than the $O(N)$ scaling of the dense Hebbian perceptron. The price is that each pattern carries very little information (only $M \ll N$ active bits), so the total information stored per synapse is more modest. But the number of distinct *patterns* is enormous.

The tradeoff is fundamental: if M is too small, each pattern is so sparse that it barely uses the network's resources, but the error rate is low. If M is too large, patterns overlap heavily and retrieval fails. The optimum sits between these extremes.

Chapter Summary

Concept	Equation	Key Insight
Hebbian rule	$\Delta w_i = \eta x_i y$	Local, correlation-based weight update
Hebbian weight vector	$\mathbf{w} = \frac{1}{N_u} \sum_m v^{(m)} \mathbf{u}^{(m)}$	Sum of input–output outer products
Retrieval	$\mathbf{w} \cdot \mathbf{u}^{(n)} = v^{(n)} + \eta^{(n)}$	Signal plus crosstalk noise
Noise variance	$\sigma_\eta^2 = \frac{N_s - 1}{N_u}$	More patterns ⇒ more interference
Sparse capacity	$P \leq \rho\left(\frac{N}{M}\right)^2$	Sparse codes store more patterns
Error criterion	$N p_{\text{on}}^M < 1$	Constrains minimum sparsity

Conceptual Questions

Q2.1 Distinguish between supervised, unsupervised, and reinforcement learning. Which category does the Hebbian rule fall into, and why?

Q2.2 The Hebbian rule $\Delta w_i = \eta\, x_i\, y$ is called a "local" learning rule. What does locality mean here, and why is it biologically important?

Q2.3 Naive Hebbian learning can cause weights to grow without bound. Why? What mechanisms could prevent this in a biological network?

Q2.4 Explain how Hebbian learning can produce an association between two previously unrelated stimuli (e.g., the sight and smell of food). Walk through the learning process step by step.

Q2.5 In the retrieval equation $\mathbf{w} \cdot \mathbf{u}^{(n)} = v^{(n)} + \eta^{(n)}$, where does the noise term $\eta^{(n)}$ come from? Decompose it and explain why its variance grows with the number of stored patterns.

Q2.6 The noise variance is $\sigma_\eta^2 = (N_s - 1)/N_u$. What does this formula tell you about the tradeoff between the number of stored patterns N_s and the input dimension N_u? Roughly how many patterns can be stored reliably?

Q2.7 Why is the central limit theorem needed to analyze the crosstalk noise? What conditions must hold for it to apply, and what does it tell us about the distribution of $\eta^{(n)}$?

Q2.8 The Hebbian rule is "not optimal" for storage capacity. What does this mean, and what property of the Hebbian rule causes it to waste capacity compared to an optimal algorithm?

Q2.9 In the sparse binary synapse model, why does making patterns sparser increase the number of storable patterns? What prevents us from making patterns arbitrarily sparse?

Q2.10 The sparse model uses binary synapses that can only be turned on, never off. What problem does this create as the number of stored patterns grows, and how does the error criterion $Np_{\text{on}}^M < 1$ address it?

Q2.11 The dense Hebbian perceptron stores $O(N)$ patterns, while the sparse binary model stores $O(N^2/\log^2 N)$ patterns. Where does this dramatic improvement come from? Is the total *information* stored per synapse also dramatically better, or is there a tradeoff?

3

Visual Receptive Fields

How does the visual cortex represent the world? This chapter develops the linear filter model of simple cells, shows why a Gabor function captures their spatial tuning, and explains how complex cells achieve phase invariance through the energy model. The key distinction is between separable and non-separable space-time receptive fields, which determines whether a neuron can detect motion direction.

3.1 Receptive Fields and the Linear Model

The *receptive field* of a visual neuron is the region of the visual field that influences its firing. Hubel and Wiesel's classic experiments showed that neurons in primary visual cortex (V1) respond selectively to oriented bars and edges at particular positions. A vertical bar might drive a neuron strongly, while a horizontal bar at the same location produces no response at all. The neuron's tuning curve—its firing rate as a function of stimulus orientation—has a clear peak at its preferred angle.

The simplest quantitative model treats the neuron as a *linear filter* acting on the visual stimulus. The stimulus $s(x, y, t)$ is a function of space and time (the light intensity at each point in the visual field at each moment). The neuron's response at time t is a weighted integral over recent stimulus history:

Linear Receptive Field Model

$$L(t) = \int_0^\infty \mathrm{d}\tau \iint \mathrm{d}x\ \mathrm{d}y\ D(x, y, \tau)\, s(x, y, t - \tau) \tag{3.1}$$

$L(t)$: linear response, $D(x, y, \tau)$: space-time receptive field kernel, s: stimulus.

This is a convolution: the kernel $D(x, y, \tau)$ tells us how much weight the neuron gives to stimulus at position (x, y) presented τ seconds in the past. Positive values of D indicate ON regions (the neuron likes light there); negative values indicate OFF regions (the neuron is suppressed by light there). The response $L(t)$ is the running correlation between the stimulus and the neuron's template.

This model is the starting point. It cannot be the whole story—firing rates are non-negative, and real neurons saturate—but it captures the essential tuning properties of simple cells remarkably well.

3.2 How Receptive Fields Are Measured

The linear-kernel model is only useful if we can estimate $D(x, y, \tau)$ from data. In practice, experiments probe the neuron with a stimulus family chosen to isolate different structure in the receptive field.

Common stimulus classes are: static gratings (orientation and phase tuning), counterphase gratings (phase sensitivity and frequency doubling), drifting gratings (direction and velocity tuning), and spatiotemporal white noise (unbiased estimation of the full kernel). Each class emphasizes a different projection of the same underlying receptive field.

For white-noise stimuli, a standard estimator is the spike-triggered average (STA):

$$\widehat{D}(x, y, \tau) \propto \left\langle s(x, y, t - \tau) \,|\, \text{spike at } t \right\rangle_t \tag{3.2}$$

If the linear-nonlinear assumptions hold and stimulus correlations are handled appropriately, the STA recovers the linear filter up to scale. This is one reason white noise is so central in sensory physiology: it turns receptive-field estimation into a controlled regression problem.

Computationally, this measurement workflow links theory to experiment. The model is not just a conceptual picture; its parameters can be inferred directly from recorded spikes and then used to predict responses to novel stimuli.

3.3 Separability of Space and Time

The full kernel $D(x, y, \tau)$ is a function of three variables, which makes it hard to measure and hard to think about. A major simplification occurs when the kernel is *separable*:

$$D(x, y, \tau) = D_s(x, y)\, D_t(\tau) \tag{3.3}$$

This means the spatial structure (which positions excite or inhibit the neuron) is independent of the temporal structure (how the neuron's sensitivity evolves over time). Many V1 simple cells are approximately separable, which lets us study spatial and temporal tuning independently.

When both the kernel and the stimulus are separable, the response factors as well. If $s(x, y, t) = s_s(x, y)\, s_t(t)$, then $L(t) = L_s \cdot L_t(t)$, where $L_s = \iint D_s\, s_s \; \mathrm{d}x \; \mathrm{d}y$ is a spatial overlap

and $L_t(t) = \int D_t(\tau)\, s_t(t-\tau)\ \mathrm{d}\tau$ is a temporal convolution. The spatial part determines *how much* the neuron responds; the temporal part determines *when*.

Separability is not universal, however. The temporal kernel of real neurons shows polarity reversal: the preferred stimulus flips from ON to OFF over roughly 50–100 ms. When plotted in the x-τ plane (suppressing y), a separable kernel has vertical or horizontal structure. A non-separable kernel has a *tilted* structure—and this tilt is the key to direction selectivity, as we will see in Section 3.5.

3.4 The Gabor Model of Simple Cells

Hubel and Wiesel found that V1 simple cells have elongated receptive fields with alternating ON and OFF subregions. The spatial kernel that captures this structure is a *Gabor function*: a sinusoidal grating modulated by a Gaussian envelope.

Gabor Spatial Receptive Field

$$D_s(x,y) = \frac{1}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{x^2}{2\sigma_x^2} - \frac{y^2}{2\sigma_y^2}\right) \cos(kx - \phi) \tag{3.4}$$

k: preferred spatial frequency, ϕ: preferred spatial phase, σ_x, σ_y: envelope widths (orientation is set by rotating coordinates).

The Gaussian envelope localizes the filter in space: the neuron only cares about a small patch of the visual field. The cosine creates alternating excitatory and inhibitory bands. Together, they produce a localized, oriented, band-pass filter.

The response to a sinusoidal grating of spatial frequency K and orientation Θ is maximal when: (1) the grating orientation matches the filter's preferred orientation, (2) the spatial frequency K matches the filter's preferred frequency k, and (3) the phase Φ matches ϕ. This explains the sharp orientation tuning and spatial frequency selectivity observed in simple cells.

The Gabor model is not just a convenient fit. It reflects a deep principle: the Gabor function achieves the minimum joint uncertainty in space and spatial frequency allowed by the uncertainty principle. A narrower Gaussian gives better spatial localization but worse frequency selectivity, and vice versa. Real simple cells sit near this optimal tradeoff.

3.5 Direction Selectivity and Non-Separability

A drifting grating moves across the visual field at a constant velocity. In the x-t plane, it appears as a tilted pattern: $s(x,t) = A\cos(Kx - \omega t)$, where the tilt angle encodes speed and direction. A grating drifting in the opposite direction has the opposite tilt: $s(x,t) = A\cos(Kx + \omega t)$.

A neuron with a separable space-time kernel $D(x,\tau) = D_s(x)\,D_t(\tau)$ cannot distinguish between leftward and rightward motion. The proof is geometric: a separable kernel has no preferred tilt in the x-τ plane, so it responds equally to both directions.

To detect motion direction, the kernel must be non-separable:

$$D(x,\tau) \neq D_s(x)\,D_t(\tau) \tag{3.5}$$

A non-separable kernel has a tilted structure in x-τ space that matches the tilt of the preferred direction. The gradient of the ON region in space-time defines the neuron's preferred velocity. This is one of the cleanest examples in neuroscience of a computational requirement (direction discrimination) dictating a structural property (non-separability).

3.6 The Static Nonlinearity

The linear model predicts a response $L(t)$ that can be negative, which is unphysical for a firing rate. It also fails to saturate at high contrast. The standard fix is to pass $L(t)$ through a static nonlinearity:

$$r(t) = r_0 + F(L(t)) \tag{3.6}$$

where r_0 is a baseline rate and F is a rectifying or sigmoidal function. This *linear–nonlinear* (LN) model preserves the tuning properties of the linear stage while producing realistic firing rates. The nonlinearity is "static" because it is applied independently at each time point, with no memory of past responses.

3.7 Complex Cells and the Energy Model

Not all V1 neurons are simple cells. *Complex cells* share the orientation selectivity of simple cells but differ in a crucial way: they are *phase invariant*. A simple cell responds best to a grating at one particular phase (say, a bright bar centered on its ON region) and poorly at the opposite phase. A complex cell responds equally well to both. It cares about the orientation and frequency of the stimulus but not about exactly where the bars fall within its receptive field.

Complex cells also show *frequency doubling*: when stimulated with a counterphase grating (one that flickers in place rather than drifting), a simple cell modulates its firing at the stimulus frequency ω, while a complex cell modulates at 2ω. This is a signature of a squaring nonlinearity.

3.7.1 The energy model

The standard model for complex cells is the *energy model*, proposed by Adelson and Bergen. Take two simple cells with identical orientation and frequency preferences but spatial phases offset by 90° (a *quadrature pair*). If their linear responses are $L_1 \propto \cos(\phi - \Phi)$ and $L_2 \propto \sin(\phi - \Phi)$, the energy is

Energy Model for Complex Cells

$$E = L_1^2 + L_2^2 \tag{3.7}$$

L_1, L_2: responses of two simple cells in quadrature (phase offset $\pi/2$).

Phase invariance follows immediately: $\cos^2\alpha + \sin^2\alpha = 1$ for any α. The energy does not depend on the stimulus phase Φ, exactly as observed in complex cells.

Frequency doubling also falls out naturally. For a counterphase grating with temporal modulation $\cos(\omega t)$, each simple cell's response contains $\cos(\omega t)$. Squaring produces $\cos^2(\omega t) = \frac{1}{2}(1 + \cos 2\omega t)$—a DC component plus a term at twice the stimulus frequency.

Hubel and Wiesel proposed a mechanistic interpretation: a complex cell pools the outputs of multiple simple cells that share the same orientation and frequency preferences but span different positions and phases within the receptive field. The energy model makes this hierarchical pooling precise.

3.8 The Visual Hierarchy

The progression from retina to V1 illustrates a recurring theme in sensory neuroscience: successive stages of filtering and pooling build increasingly abstract representations.

Retinal ganglion cells and LGN neurons have center-surround receptive fields—roughly circular, with an excitatory center and inhibitory surround (or vice versa). These are not orientation selective; they respond to local contrast. V1 simple cells, by combining inputs from multiple LGN neurons with appropriately aligned receptive fields, create oriented filters.

Complex cells, by pooling over simple cells, gain phase invariance. Each stage discards some information (phase, exact position) and preserves what matters for the next level of processing (orientation, spatial frequency, motion direction).

This hierarchy—from local contrast to oriented edges to phase-invariant features—is the computational backbone of early vision. Higher areas continue the pattern, building selectivity for texture, shape, and ultimately objects and faces, but those stages are beyond the scope of this chapter.

Chapter Summary

Concept	Equation	Key Insight
Linear RF model	$L(t) = \iint D\,s\ \mathrm{d}x\ \mathrm{d}y\ \mathrm{d}\tau$	Neuron as a linear filter (template matching)
Separability	$D(x, y, \tau) = D_s D_t$	Space and time decouple
Gabor filter	Gaussian × cosine	Oriented, localized, band-pass spatial filter
Non-separability	$D(x, \tau) \neq D_s D_t$	Required for direction selectivity
LN model	$r(t) = r_0 + F(L(t))$	Rectifying nonlinearity fixes negative rates
Energy model	$E = L_1^2 + L_2^2$	Quadrature pair ⇒ phase invariance

Conceptual Questions

Q3.1 The linear receptive field model $L(t) = \iint D\,s\ \mathrm{d}x\ \mathrm{d}y\ \mathrm{d}\tau$ is a convolution. In plain language, what is this operation doing? Why is it sometimes called "template matching"?

Q3.2 What is the difference between the classical and non-classical receptive field of a neuron? How might the non-classical surround affect the neuron's response to a stimulus inside its classical receptive field?

Q3.3 Explain what it means for a space-time receptive field to be *separable*. If $D(x, y, \tau) = D_s(x, y)\, D_t(\tau)$, how does the response to a stimulus factor into spatial and temporal components?

Q3.4 A simple cell has a separable space-time receptive field. A drifting grating and its reverse (same speed, opposite direction) are presented. Why does the cell respond identically to both? What would need to change for it to prefer one direction?

Q3.5 The Gabor filter has parameters k (spatial frequency), ϕ (phase), and σ_x, σ_y (envelope widths). Describe what each parameter controls about the neuron's selectivity. What happens to spatial frequency selectivity if you narrow the Gaussian envelope?

Q3.6 Why does a Gabor filter act as a band-pass filter in spatial frequency? What is the role of the Gaussian envelope versus the cosine component?

Q3.7 A simple cell responds to a grating at its preferred orientation, spatial frequency, and phase. What happens to the response if you shift the grating by half a spatial period (a phase shift of π)? How does this compare to a complex cell?

Q3.8 Why is the linear model insufficient on its own for predicting firing rates? What does the static nonlinearity F need to do, and why is it called "static"?

Q3.9 The energy model predicts frequency doubling in complex cells. Show, using $\cos^2(\omega t) = \frac{1}{2}(1 + \cos 2\omega t)$, why the response to a counterphase grating modulates at twice the stimulus frequency.

Q3.10 Hubel and Wiesel proposed that complex cells pool over simple cells with the same orientation preference but different phases. How does this relate to the mathematical structure of the energy model $E = L_1^2 + L_2^2$?

Q3.11 Describe the visual hierarchy from retina to V1. What information is discarded at each stage, and what is preserved? Why is this progressive abstraction computationally useful?

Q3.12 The temporal kernel of a V1 simple cell typically shows polarity reversal: the preferred polarity flips after roughly 50–100 ms. What does this reversal look like in the x-τ plot, and what is its functional significance?

Part II

Biophysics of Single Neurons

4

The Membrane Potential

Every neuron maintains a voltage difference across its membrane—roughly -70 mV at rest. This chapter explains why this voltage exists, how we can predict it from first principles, and what happens when we inject current into a cell. The key ideas are diffusion, drift, the Nernst equation, and the membrane as an RC circuit.

4.1 The Point Neuron Approximation

We begin with a drastic simplification: treat the entire neuron as a single electrical compartment. The membrane is an insulating lipid bilayer separating two conductive solutions. We ignore the spatial extent of dendrites and axons and ask: what determines the voltage V across this membrane? Relaxing this assumption leads to cable theory (Chapter 11).

The key physical insight is that the lipid bilayer is an insulator sandwiched between two conductors (intra- and extracellular fluid). This is precisely a capacitor. The membrane stores charge, and the voltage across it is determined by the balance of ionic currents flowing through channels and the charge stored on the capacitor.

The fundamental experimental observation is this: if you inject a small current into a neuron, the voltage deflects smoothly from rest. But above a threshold, the response is dramatically different—a sharp, stereotyped spike. This *all-or-nothing response* is the action potential. Understanding why requires understanding the ionic basis of the resting potential.

4.2 Ions and Concentrations

The relevant charge carriers are ions dissolved in the intra- and extracellular fluid. The key players are:

Ion	**Valence** z	**Inside** (mM)	**Outside** (mM)
Na^+	$+1$	5–15	~ 145
K^+	$+1$	~ 140	~ 5
Cl^-	-1	~ 4	~ 110
Ca^{2+}	$+2$	~ 0.1 μM	2.5–5

These concentrations are maintained by active transport, particularly the Na^+/K^+-ATPase pump, which expends energy to move $3\,Na^+$ out and $2\,K^+$ in per ATP hydrolyzed.

A crucial observation: the bulk solutions on either side of the membrane are very nearly electrically neutral—a property called *space-charge neutrality*. The total positive charge approximately equals the total negative charge in both compartments. The membrane potential arises from a *thin layer* of excess charge at the membrane surface—a tiny fraction of the total ions present.

The Faraday constant $F = N_A q_e \approx 96{,}485\ \mathrm{C\,mol^{-1}}$ converts between moles and coulombs. One mole of monovalent ions carries F coulombs of charge.

4.3 Diffusion: Fick's Law

Imagine dropping a blob of dye into water. The dye molecules spread out—not because any force pushes them, but because random thermal motion carries more molecules away from regions of high concentration than toward them. This is diffusion. Each ion undergoes a random walk due to thermal collisions. On average, there is no preferred direction, but in a region with a concentration gradient, more particles randomly step *away* from the dense region than *toward* it, producing a net flux from high to low concentration.

We can derive the diffusion flux from a simple random walk model. Consider a one-dimensional lattice with spacing Δx. At each time step Δt, each particle steps left or right with equal probability 1/2. If the concentration at position x is $C(x)$, the net flux across a boundary is

Fick's Law

$$J_{\mathrm{diff}} = -D\frac{\partial C}{\partial x} \tag{4.1}$$

J_{diff}: particle flux ($\mathrm{mol\,m^{-2}s^{-1}}$), D: diffusion coefficient ($\mathrm{m^2/s}$), C: concentration ($\mathrm{mol/m^3}$)

where the diffusion coefficient $D = (\Delta x)^2/(2\Delta t)$. The negative sign captures the essential physics: particles flow *down* the concentration gradient, from high to low concentration.

4.4 Drift in an Electric Field

Ions carry charge, so they respond to electric fields. An electric field $E = -\,\mathrm{d}U/\,\mathrm{d}x$ exerts a force on each ion, producing a directed drift on top of the random thermal motion. The drift flux for ions of valence z and concentration C is

$$J_{\text{drift}} = -\mu z C \frac{\mathrm{d}U}{\mathrm{d}x} \tag{4.2}$$

where μ is the *mobility*—how fast ions move per unit force, accounting for friction with the surrounding medium.

Diffusion and drift are not independent phenomena. Both arise from the same thermal fluctuations and friction. The *Einstein relation* connects them:

$$D = \mu \frac{k_B T}{q} = \mu \frac{RT}{zF} \tag{4.3}$$

Hotter temperatures mean faster diffusion *and* faster drift for the same applied force. This relation is the key to deriving the Nernst equation.

4.5 The Nernst Equation

We now ask: at what voltage does the net ionic current vanish for a single ionic species? This is the equilibrium condition where diffusion exactly balances drift.

4.5.1 Derivation from zero net flux

The total flux is the sum of diffusion and drift:

$$J = -D \frac{\partial C}{\partial x} - \mu z C \frac{\mathrm{d}U}{\mathrm{d}x} \tag{4.4}$$

At equilibrium, $J = 0$. Setting the two terms equal and using the Einstein relation $D = \mu RT/(zF)$:

NERNST EQUATION

Start from $J = 0$:

$$D \frac{\partial C}{\partial x} = -\mu z C \frac{\mathrm{d}U}{\mathrm{d}x}$$

$$\frac{RT}{zF} \frac{1}{C} \frac{\partial C}{\partial x} = -\frac{\mathrm{d}U}{\mathrm{d}x} \qquad \text{substituting } D = \frac{\mu RT}{zF}$$

The left side is $\frac{RT}{zF}\frac{\mathrm{d}}{\mathrm{d}x}\ln C$. Integrating across the membrane from inside ($x = 0$) to outside ($x = d$):

$$\frac{RT}{zF}\ln\frac{C_{\text{out}}}{C_{\text{in}}} = -(U_{\text{out}} - U_{\text{in}}) = V$$

where $V = U_{\text{in}} - U_{\text{out}}$ is the membrane potential (intracellular minus extracellular).

The Nernst Equation

$$E_{\text{X}} = \frac{RT}{zF}\ln\frac{[\text{X}]_{\text{out}}}{[\text{X}]_{\text{in}}} \tag{4.5}$$

$R = 8.314\ \text{J mol}^{-1}\text{K}^{-1}$: gas constant, T: temperature (K), z: ion valence, $F = 96{,}485\ \text{C mol}^{-1}$: Faraday constant.

At body temperature $T = 310$ K: $RT/F \approx 26.7$ mV.

The quantity E_{ion} is the voltage at which the electrical force on an ion exactly balances its tendency to diffuse down its concentration gradient. If the membrane potential equals the Nernst potential for a given ion, that ion has zero net current—even if channels for that ion are open.

4.5.2 The concept of a reversal potential

The Nernst potential E_{ion} is also called the *reversal potential* because it is the voltage where the ionic current reverses direction. If $V > E_{\text{ion}}$, the driving force ($V - E_{\text{ion}}$) is positive and a cation channel conducts outward current. If $V < E_{\text{ion}}$, the driving force is negative and the channel conducts inward current. At $V = E_{\text{ion}}$, no net current flows.

As a concrete example, using $RT/F \approx 26.7$ mV at body temperature: for potassium with $[\text{K}]_{\text{out}} = 5$ mM and $[\text{K}]_{\text{in}} = 140$ mM,

$$E_{\text{K}} = 26.7\ \text{mV} \times \ln\frac{5}{140} \approx -89\ \text{mV}, \tag{4.6}$$

$$E_{\text{Na}} = 26.7\ \text{mV} \times \ln\frac{145}{10} \approx +71\ \text{mV}. \tag{4.7}$$

The sign tells us the direction of the equilibrium force: K^+ wants to pull the cell negative; Na^+ wants to push it positive. At rest, the membrane is much more permeable to K^+, so the resting potential (≈ -70 mV) is close to E_{K}.

Note that the Nernst equation gives the equilibrium potential for a *single* ionic species, assuming no other ions are permeant. A real membrane is permeable to multiple ions simultaneously, which is why the resting potential is not exactly equal to any single Nernst potential.

4.6 Multiple Ions: The Goldman–Hodgkin–Katz Equation

A real membrane has channels for several ion types simultaneously. The resting potential depends on all of them, weighted by their permeabilities. The Goldman–Hodgkin–Katz (GHK) voltage equation gives the voltage at which the *total* ionic current vanishes. It assumes a constant electric field across the membrane, non-interacting ions, and steady state (not equilibrium—ions are still flowing, but the net current is zero).

4.6.1 Donnan equilibrium versus physiological steady state

It is important to separate two concepts that are often conflated. In a true thermodynamic equilibrium (Donnan equilibrium), all ionic fluxes vanish individually and each permeant ion satisfies its own Nernst condition at the same membrane voltage. Real neurons are not in that state. They maintain concentration gradients by active transport (Na/K pump, exchangers), so there are ongoing ionic fluxes and metabolic work.

Resting neurons are better described as a *nonequilibrium steady state*: concentrations are approximately time-invariant, and the *net membrane current* is zero, but individual ionic currents need not vanish. The GHK framework is built precisely for this regime.

Goldman–Hodgkin–Katz Voltage Equation

$$V_{\text{rest}} = \frac{RT}{F} \ln \frac{P_{\text{K}}[\text{K}]_{\text{out}} + P_{\text{Na}}[\text{Na}]_{\text{out}} + P_{\text{Cl}}[\text{Cl}]_{\text{in}}}{P_{\text{K}}[\text{K}]_{\text{in}} + P_{\text{Na}}[\text{Na}]_{\text{in}} + P_{\text{Cl}}[\text{Cl}]_{\text{out}}} \tag{4.8}$$

P_X: permeability of ion X (cm/s). Cl^- concentrations are flipped (in/out) because $z_{\text{Cl}} = -1$.

The GHK equation is effectively a permeability-weighted average of the individual Nernst potentials. Whichever ion has the highest permeability dominates the resting potential. At rest, $P_{\text{K}} \gg P_{\text{Na}}$, so $V_{\text{rest}} \approx E_{\text{K}} \approx -70$ mV.

4.6.2 GHK current equation and nonlinear I–V curves

The voltage equation above is derived by summing GHK *currents*. For a single ion species X with valence z_X under the constant-field approximation:

Goldman–Hodgkin–Katz Current Equation

$$I_X(V) = P_X \frac{z_X^2 F^2}{RT} V \frac{[\mathrm{X}]_{\mathrm{in}} - [\mathrm{X}]_{\mathrm{out}} \exp(-z_X FV/(RT))}{1 - \exp(-z_X FV/(RT))} \tag{4.9}$$

P_X: permeability, z_X: valence, F, R, T: Faraday constant, gas constant, temperature.

Unlike an Ohmic current, Equation (4.9) is nonlinear in V. This nonlinearity can produce inward or outward rectification even for a passive membrane with fixed permeabilities, purely from electrodiffusion.

At small voltages ($|z_X FV| \ll RT$), expanding the exponential recovers an approximately linear relation $I_X \approx g_X(V - E_X)$, which explains why Ohmic channel models are often reasonable near rest.

4.6.3 An alternative passive nonlinearity: energy barriers

A second route to non-Ohmic behavior is an internal energy barrier in the channel pore (Eyring rate theory). In the simplest one-barrier model, inward and outward hopping rates depend asymmetrically on membrane voltage:

$$I \propto [\mathrm{X}]_{\mathrm{in}} \exp\Big(-\delta \frac{zFV}{RT}\Big) - [\mathrm{X}]_{\mathrm{out}} \exp\Big((1-\delta)\frac{zFV}{RT}\Big), \tag{4.10}$$

where $\delta \in [0, 1]$ is the fractional barrier position within the membrane field. When $\delta = 1/2$ the barrier is symmetric; $\delta \neq 1/2$ yields asymmetric rectification.

The key conceptual point is that passive nonlinear I–V curves need not imply voltage-gated channel dynamics. They can arise from electrodiffusion (GHK) or from asymmetric barrier crossing (Eyring) even before active gating is introduced.

Computationally, this matters because the baseline membrane transfer function may already be nonlinear before we add voltage-dependent channels, altering gain and subthreshold filtering.

4.6.4 Why extracellular potassium depolarizes neurons

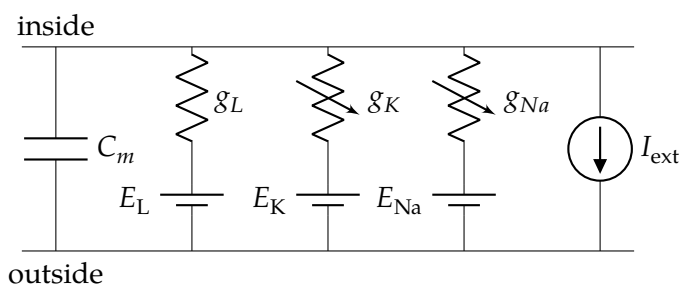
If we increase $[K]_{out}$ (a common clinical scenario in hyperkalemia), the Nernst potential E_K becomes less negative because the log ratio $[K]_{out}/[K]_{in}$ increases. Since the resting potential is dominated by K^+ permeability, the resting potential moves toward zero—the cell depolarizes. For example, raising $[K]_{out}$ from the normal 5 mM to 15 mM shifts E_K from −89 mV to roughly −60 mV, a significant depolarization that can make neurons hyperexcitable.

4.7 The Membrane as an RC Circuit

We now model the membrane as an electrical circuit. This is the foundation for every dynamical neuron model in this course.

4.7.1 Circuit components

The membrane has three electrical properties: a **capacitance** C_m (the lipid bilayer stores charge; typical value $\sim 1\ \mu\text{F/cm}^2$), **conductances** g_i (ion channels act as resistors, each with its own reversal potential), and an **external current** I_{ext} (injected by an electrode or, physiologically, by synapses).



4.7.2 Kirchhoff's current law

By Kirchhoff's law, the total current into the cell must equal the sum of all currents leaving through channels plus the current charging the capacitor. We adopt the sign convention that positive current flows *into* the cell, and ionic currents are positive when flowing outward.

The Membrane Equation

$$C_m \frac{\mathrm{d}V}{\mathrm{d}t} = -\sum_i g_i(V - E_\mathrm{i}) + I_\mathrm{ext}(t) \tag{4.11}$$

C_m: membrane capacitance, g_i: conductance of channel type i, E_i: reversal potential, I_ext: injected current.

Each term has a clear physical meaning. The left side $C_m\,\mathrm{d}V/\mathrm{d}t$ is the rate of charge accumulation on the capacitor. Each $-g_i(V - E_\mathrm{i})$ term is the current through channel type i; the factor $(V - E_\mathrm{i})$ is the *driving force*, which vanishes at the reversal potential, reverses sign across it, and grows linearly away from it. The term I_ext is whatever current we inject.

4.7.3 Single-conductance case: the passive membrane

The simplest case has a single leak conductance g_L with reversal potential $E_\mathrm{L} \approx V_\mathrm{rest}$:

$$C_m \frac{\mathrm{d}V}{\mathrm{d}t} = -g_L(V - E_\mathrm{L}) + I_\mathrm{ext}(t) \tag{4.12}$$

This defines the membrane time constant and input resistance:

$$\tau_m = \frac{C_m}{g_L}, \qquad R_m = \frac{1}{g_L} \tag{4.13}$$

For typical values $C_m = 200$ pF and $g_L = 10$ nS, we get $\tau_m = 20$ ms and $R_m = 100$ MΩ. The time constant sets the temporal scale of subthreshold voltage dynamics.

4.7.4 Response to a constant current step

Suppose $V(0) = E_\mathrm{L}$ and we inject a constant current I_0 starting at $t = 0$. Dividing Equation (4.12) by g_L:

$$\tau_m \frac{\mathrm{d}V}{\mathrm{d}t} = -(V - E_\mathrm{L}) + R_m I_0 \tag{4.14}$$

This is a first-order linear ODE. The solution is an exponential approach to a new steady state:

$$V(t) = E_\mathrm{L} + R_m I_0 \left(1 - e^{-t/\tau_m}\right) \tag{4.15}$$

The voltage rises exponentially from E_L toward the steady-state value $V_\infty = E_\mathrm{L} + R_m I_0$ with time constant τ_m. This exponential charging is precisely the subthreshold dynamics

of the leaky integrate-and-fire neuron (Chapter 5). Adding a spike threshold and reset to Equation (4.12) gives the LIF model—the simplest spiking neuron. The membrane time constant τ_m sets how long the neuron "remembers" its input.

Note that adding more open channels *decreases* the time constant: the membrane responds faster but the voltage excursions become smaller, since $R_{\text{eff}} = 1/g_{\text{tot}}$ decreases. Inhibition with $E_{\text{inh}} \approx V_{\text{rest}}$ does not hyperpolarize the cell, but it "shunts" excitatory drive by dividing the input resistance. This is called *shunting inhibition*.

Chapter Summary

Concept	Equation	Key Insight
Fick's law	$J_{\text{diff}} = -D\,\dfrac{\partial C}{\partial x}$	Particles flow down concentration gradient
Einstein relation	$D = \dfrac{\mu RT}{zF}$	Links diffusion to drift
Nernst equation	$E_X = \dfrac{RT}{zF}\ln\dfrac{[\text{X}]_{\text{out}}}{[\text{X}]_{\text{in}}}$	Zero-current voltage for one ion
GHK equation	$V = \dfrac{RT}{F}\ln\dfrac{P_K[\text{K}]_{\text{out}} + \cdots}{P_K[\text{K}]_{\text{in}} + \cdots}$	Permeability-weighted multi-ion potential
Membrane equation	$C_m\dot{V} = -\sum g_i(V - E_{\text{i}}) + I_{\text{ext}}$	RC circuit with ionic batteries
Time constant	$\tau_m = C_m/g_{\text{tot}}$	Timescale of voltage dynamics
Resting potential	$V_{\text{rest}} = \dfrac{\sum g_i E_{\text{i}}}{\sum g_i}$	Conductance-weighted average

Conceptual Questions

Q4.1 The lipid bilayer is an insulator between two conductors. Why does this make it a capacitor? What determines the capacitance?

Q4.2 Diffusion arises from random thermal motion, yet it produces a directed net flux. Explain this apparent paradox using the concept of a concentration gradient.

Q4.3 Why does Fick's law have a negative sign? What would happen physically if the sign were positive?

Q4.4 The Einstein relation $D = \mu k_B T/q$ links diffusion and drift. What does it mean physically that both depend on temperature?

Q4.5 Why does the Nernst equation have a logarithm? What physical balance does it represent? Walk through the derivation from setting the total flux to zero.

Q4.6 The Nernst potential for K^+ is approximately -89 mV and for Na^+ is approximately $+71$ mV. Explain the sign of each in terms of the concentration gradients.

Q4.7 The resting potential of a typical neuron (≈ -70 mV) is close to E_K but not exactly equal. Why not?

Q4.8 A patient has dangerously elevated blood potassium ($[K]_{out} = 8$ mM instead of 5 mM). Without computing exact numbers, explain qualitatively what happens to the resting potential and why this is dangerous.

Q4.9 Why is the Nernst equation an *equilibrium* result while the GHK equation describes a *steady state*? What is the difference?

Q4.10 In the GHK equation, the Cl^- concentrations appear with inside and outside reversed compared to the cations. Why? What would happen if you forgot to flip them?

Q4.11 In the membrane equation, what is the physical meaning of each term: $C_m\, dV/dt$, $-g_i(V - E_i)$, and I_{ext}?

Q4.12 The driving force $(V - E_i)$ can be positive, negative, or zero. What does each case correspond to physically? What happens to the current through channel type i when $V = E_i$?

Q4.13 In the membrane equation, what happens to the voltage dynamics if you double all conductances simultaneously? Does the resting potential change? Does the time constant change?

Q4.14 The step response of a passive membrane is $V(t) = E_L + R_m I_0(1 - e^{-t/\tau_m})$. Explain qualitatively why the voltage approaches a new steady state exponentially rather than linearly. What determines the final value?

Q4.15 Explain shunting inhibition in terms of the membrane equation. Why does adding an inhibitory conductance with $E_{inh} \approx V_{rest}$ reduce the response to excitation without hyperpolarizing the cell?

5

Ion Channels, Synapses, and the Integrate-and-Fire Neuron

The membrane equation from Chapter 4 describes how voltage evolves, but it says nothing about what makes conductances change or what happens when voltage reaches threshold. This chapter introduces ion channels as the molecular basis of conductance, develops the leaky integrate-and-fire model as the simplest spiking neuron, and describes how synaptic inputs drive the membrane.

5.1 Ion Channels

Ion channels are protein pores in the cell membrane that allow specific ions to cross. Each channel has two essential properties: *selectivity* (which ions can pass through) and *gating* (what controls whether the channel is open or closed). The current through a population of channels of type i is

$$I_i = g_i(V, t)(V - E_{\mathrm{i}}) \tag{5.1}$$

where $g_i(V, t)$ is the conductance—the total number of open channels times the conductance per channel—and E_{i} is the reversal potential for that ion species. The factor $(V - E_{\mathrm{i}})$ is the driving force: current flows inward when $V < E_{\mathrm{i}}$ and outward when $V > E_{\mathrm{i}}$.

Different channels are gated by different stimuli. *Voltage-gated channels* open and close in response to changes in membrane potential. These are the channels responsible for the action potential, and we will analyze their gating kinetics in detail in Chapter 6. Other channels are gated by ligands (neurotransmitters), by mechanical force, or by temperature.

The key point for now is that when channels are voltage-gated, the conductance g_i depends on V, which in turn depends on g_i through the membrane equation. This creates *feedback*. Depolarization can open channels that allow inward current, which further depolarizes the membrane—a positive feedback loop that underlies the explosive upstroke of the action potential.

5.2 Voltage Clamp

The voltage clamp is the experimental technique that made quantitative channel biophysics possible. The idea is to hold the membrane voltage at a fixed value V_{cmd} using an electronic feedback circuit, and measure the current $I(t)$ required to maintain that voltage. Since V is constant, $C_m \, \mathrm{d}V/\mathrm{d}t = 0$, and the measured current equals the total ionic current:

$$I_{\text{clamp}}(t) = \sum_i g_i(V_{\text{cmd}}, t)(V_{\text{cmd}} - E_{\text{i}}) \tag{5.2}$$

By stepping V_{cmd} to different values and using pharmacological blockers to isolate individual channel types, experimenters can extract the voltage dependence and kinetics of each conductance. This is how Hodgkin and Huxley determined the gating variables of the squid giant axon (Chapter 6).

Without voltage clamp, the feedback between voltage and conductance makes the system nearly impossible to analyze: changing the voltage changes the conductance, which changes the current, which changes the voltage. Clamping the voltage breaks this loop and lets you study conductance in isolation.

5.3 The Leaky Integrate-and-Fire Model

The action potential is a complex, nonlinear event involving the coordinated opening and closing of multiple channel types. If we are not interested in the biophysical details of the spike itself—only in *when* the neuron fires—we can replace the spike mechanism with a simple rule: when V reaches a threshold V_{th}, the neuron "fires" (we record a spike) and V is reset to V_{reset}. Between spikes, the voltage obeys the passive membrane equation.

This is the *leaky integrate-and-fire* (LIF) model:

Leaky Integrate-and-Fire Neuron

$$\tau_m \frac{\mathrm{d}V}{\mathrm{d}t} = -(V - E_{\text{L}}) + R_m I(t) \tag{5.3}$$

$\tau_m = C_m/g_L$: membrane time constant, $R_m = 1/g_L$: input resistance, E_{L}: leak reversal potential.

If $V \geq V_{\text{th}}$: record spike, set $V \leftarrow V_{\text{reset}}$, wait τ_{ref}.

The "leaky" part is the term $-(V - E_{\text{L}})$, which pulls the voltage back toward rest. Without it, the model would be a perfect integrator: every input would accumulate forever, and the

neuron would eventually fire no matter how weak the stimulus. The leak sets a time window for integration—inputs must arrive within roughly τ_m of each other to sum effectively.

After each spike, the neuron enters a *refractory period* of duration τ_{ref} during which it cannot fire again. This sets a maximum firing rate of $1/\tau_{\text{ref}}$.

5.3.1 Firing rate under constant current

For a constant injected current I_0, we can compute the LIF firing rate exactly. Between spikes, the voltage rises exponentially from V_{reset} toward the steady state $V_\infty = E_{\text{L}} + R_m I_0$. If $V_\infty > V_{\text{th}}$, the voltage reaches threshold after a time

$$T_{\text{ISI}} = \tau_{\text{ref}} + \tau_m \ln\left(\frac{R_m I_0 - (V_{\text{reset}} - E_{\text{L}})}{R_m I_0 - (V_{\text{th}} - E_{\text{L}})}\right)$$

where T_{ISI} is the interspike interval. The firing rate is $r = 1/T_{\text{ISI}}$:

LIF Firing Rate

$$r = \left[\tau_{\text{ref}} + \tau_m \ln\left(\frac{I_0 - g_L(V_{\text{reset}} - E_{\text{L}})}{I_0 - g_L(V_{\text{th}} - E_{\text{L}})}\right)\right]^{-1} \tag{5.4}$$

$r = 0$ if $I_0 \leq g_L(V_{\text{th}} - E_{\text{L}})$ (subthreshold).

LIF FIRING RATE

Between spikes, the voltage satisfies Equation (5.3) with $I(t) = I_0$ constant. The solution starting from $V(0) = V_{\text{reset}}$ is

$$V(t) = E_{\text{L}} + R_m I_0 - (E_{\text{L}} + R_m I_0 - V_{\text{reset}})e^{-t/\tau_m}$$

Setting $V(T) = V_{\text{th}}$ and solving for T:

$$\begin{aligned} V_{\text{th}} &= E_{\text{L}} + R_m I_0 - (E_{\text{L}} + R_m I_0 - V_{\text{reset}})e^{-T/\tau_m} \\ e^{-T/\tau_m} &= \frac{E_{\text{L}} + R_m I_0 - V_{\text{th}}}{E_{\text{L}} + R_m I_0 - V_{\text{reset}}} \\ T &= \tau_m \ln\left(\frac{R_m I_0 - (V_{\text{reset}} - E_{\text{L}})}{R_m I_0 - (V_{\text{th}} - E_{\text{L}})}\right) \end{aligned}$$

Adding the refractory period and inverting gives Equation (5.4). The argument of the logarithm is positive only when $I_0 > g_L(V_{\text{th}} - E_{\text{L}})$—below this current, the steady-state voltage never reaches threshold.

The *f–I* curve (firing rate versus current) starts at zero for subthreshold currents, rises steeply at threshold, and saturates at $1/\tau_{\text{ref}}$ for large currents. The LIF model produces a continuous onset of firing—the rate can be made arbitrarily small by bringing the current just above threshold. This is a hallmark of Type I excitability, which we will formalize in terms of bifurcations in Chapter 10.

5.4 Synaptic Input

Neurons receive input not from experimenters' electrodes but from other neurons, via synapses. There are two types. *Electrical synapses* (gap junctions) are direct ionic connections between cells—fast, bidirectional, and relatively simple. *Chemical synapses* are the dominant form in the brain: a presynaptic spike triggers neurotransmitter release, which opens receptor channels on the postsynaptic membrane.

Chemical synapses come in two flavors. *Ionotropic* receptors are ligand-gated channels that open directly when the transmitter binds—they are fast (millisecond timescale). *Metabotropic* receptors trigger intracellular signaling cascades—they are slower but can modulate the neuron's behavior in more complex ways.

5.4.1 Conductance-based synapse model

The current through a synapse follows the same conductance-based form as any ion channel:

$$I_{\text{syn}}(t) = g_{\text{syn}}(t)(V - E_{\text{syn}}) \tag{5.5}$$

where $g_{\text{syn}}(t) = \bar{g}\, s(t)$ is the synaptic conductance and $s(t)$ is a gating variable driven by presynaptic spikes. Common choices for $s(t)$ include an instantaneous jump followed by exponential decay (the "exponential synapse"), a difference of two exponentials (fast rise, slower decay), or the alpha function $s(t) = (t/\tau_s)\, e^{-t/\tau_s}$.

The reversal potential E_{syn} determines whether the synapse is excitatory or inhibitory. Excitatory synapses (glutamatergic, with $E_{\text{syn}} \approx 0$ mV) drive the membrane toward depolarization. Inhibitory synapses (GABAergic, with $E_{\text{syn}} \approx -70$ to -80 mV) hold the membrane near or below rest.

5.4.2 Shunting inhibition

An interesting case arises when E_{inh} is close to the resting potential. Opening inhibitory channels does not hyperpolarize the cell much—the driving force $(V - E_{\text{inh}})$ is small near rest. But it does increase the total membrane conductance g_{tot}, which has two consequences.

First, the input resistance $R_{\text{eff}} = 1/g_{\text{tot}}$ decreases, so the same excitatory current produces a smaller voltage deflection. The inhibition acts *divisively*—it scales down the response to all inputs rather than subtracting a fixed amount. Second, the membrane time constant $\tau_{\text{eff}} = C_m/g_{\text{tot}}$ decreases, so the neuron "forgets" its inputs faster. Both effects reduce excitability without changing the resting potential.

This is *shunting inhibition*, and it is fundamentally different from hyperpolarizing inhibition (where the reversal potential is well below rest). Shunting inhibition is a gain control mechanism: it divides the neuron's response by a factor that depends on the strength of inhibition.

5.4.3 Synaptic unreliability and quantal release

Chemical transmission is probabilistic: a presynaptic spike does not always trigger vesicle release. A minimal model treats each spike as a Bernoulli trial:

$$b_k \sim \text{Bernoulli}(p_k), \qquad g_{\text{syn}}(t) = \bar{g} \sum_k b_k \, \alpha(t - t_k), \tag{5.6}$$

where p_k is release probability at spike k and
alpha is the postsynaptic conductance kernel. Even with identical presynaptic spike trains, the postsynaptic response varies from trial to trial because b_k is random.

This quantal stochasticity is not a small correction: at many central synapses, release probabilities can be well below one. The resulting variability contributes substantially to membrane-potential fluctuations and to response variability in cortical neurons.

5.4.4 Short-term facilitation and depression

Release probability and available vesicle resources change on fast timescales. A standard minimal model uses a utilization variable u (facilitation) and a resource variable R (depression):

$$\dot{u} = \frac{U - u}{\tau_f} + U(1 - u) \sum_k \delta(t - t_k), \tag{5.7}$$

$$\dot{R} = \frac{1 - R}{\tau_d} - uR \sum_k \delta(t - t_k), \tag{5.8}$$

with effective release amplitude proportional to uR at each spike. Rapid presynaptic spiking can increase u (facilitation) or deplete R (depression), depending on parameters.

Functionally, facilitation can transiently amplify weak inputs in bursts, while depression can normalize sustained high-rate input. These are dynamical filters implemented at the synapse itself, before postsynaptic integration.

An important edge case is that synaptic input need not always increase total conductance. Depending on receptor pathways and channel targets, synaptic activation can indirectly *decrease* a membrane conductance (for example by closing tonic channels), shifting both gain and effective time constant in the opposite direction.

5.4.5 Massive synaptic input and the high-conductance state

In vivo, neurons receive thousands of excitatory and inhibitory spike trains simultaneously. Averaging over many synapses yields large mean conductances plus residual fluctuations:

$$C_m \dot{V} = -g_L(V - E_\mathrm{L}) - \bar{g}_E(V - E_\mathrm{E}) - \bar{g}_I(V - E_\mathrm{I}) + \sigma_I \eta(t). \tag{5.9}$$

The effective time constant becomes

$$\tau_\mathrm{eff} = \frac{C_m}{g_L + \bar{g}_E + \bar{g}_I}, \tag{5.10}$$

which is typically much smaller than in vitro values. The membrane is therefore in a fast, fluctuation-driven regime close to threshold, where spiking can be governed by variance as much as by mean input.

Computationally, the high-conductance state explains why cortical neurons can respond quickly, show irregular firing, and remain sensitive to coincident inputs despite large ongoing synaptic bombardment.

Chapter Summary

Concept	Equation	Key Insight
Ion channel current	$I_i = g_i(V,t)(V - E_\mathrm{i})$	Conductance × driving force
Voltage clamp	$I_\mathrm{clamp} = \sum g_i(V - E_\mathrm{i})$	Fix V, measure I to isolate channels
LIF model	$\tau_m \dot{V} = -(V - E_\mathrm{L}) + R_m I$	Passive membrane + threshold/reset
LIF firing rate	$r = \left[\tau_\mathrm{ref} + \tau_m \ln \frac{\cdots}{\cdots}\right]^{-1}$	Continuous onset, saturates at $1/\tau_\mathrm{ref}$
Synaptic current	$I_\mathrm{syn} = g_\mathrm{syn}(V - E_\mathrm{syn})$	Conductance-based, reversal sets sign
Shunting inhibition	$R_\mathrm{eff} = 1/g_\mathrm{tot}$	Divisive gain control near rest
Quantal release	$b_k \sim \mathrm{Bernoulli}(p_k)$	Synaptic transmission is intrinsically stochastic
Short-term plasticity	$\dot{u}, \dot{R}$ with spike-triggered jumps	Synapse acts as a dynamic temporal filter
High-conductance state	$\tau_\mathrm{eff} = C_m/(g_L + \bar{g}_E + \bar{g}_I)$	Fast, fluctuation-driven membrane dynamics

Conceptual Questions

Q5.1 Ion channels have two key properties: selectivity and gating. Explain what each means and why both are needed for the membrane to generate an action potential.

Q5.2 The current through a channel is $I = g(V,t)(V - E)$. When voltage-gated channels are present, g depends on V, which depends on g. Explain how this creates positive feedback and why it is essential for the action potential.

Q5.3 Why does the voltage clamp break the feedback loop between voltage and conductance? What would happen if you tried to study voltage-gated channels without clamping?

Q5.4 In the LIF model, what does the "leaky" part refer to? What would the model predict if the leak were removed (a perfect integrator)? Why is this unrealistic?

Q5.5 The LIF model has no mechanism for generating the spike itself—the spike is an instantaneous event imposed by a rule. What aspects of real neural behavior does this miss, and when might it matter?

Q5.6 Explain why the LIF neuron cannot fire if $I_0 < g_L(V_{\text{th}} - E_{\text{L}})$. What is the physical meaning of this threshold current?

Q5.7 Sketch the *f–I* curve of the LIF model. Why does the firing rate start from zero at threshold? Why does it saturate at $1/\tau_{\text{ref}}$ for large currents?

Q5.8 Derive the LIF interspike interval in words: starting from V_{reset}, the voltage rises exponentially toward a steady state. Under what condition does it reach V_{th}? What determines how long this takes?

Q5.9 Compare electrical and chemical synapses. What are the advantages of each for neural computation?

Q5.10 Explain the difference between ionotropic and metabotropic receptors in terms of timescale and mechanism.

Q5.11 A conductance-based synapse has current $I_{\text{syn}} = g_{\text{syn}}(V - E_{\text{syn}})$. Why does the reversal potential E_{syn} determine whether the synapse is excitatory or inhibitory? What happens to the synaptic current as V approaches E_{syn}?

Q5.12 Shunting inhibition with $E_{\text{inh}} \approx V_{\text{rest}}$ does not change the resting potential. Why, then, does it reduce the neuron's response to excitatory input? Explain using both the input resistance and the time constant.

Q5.13 Compare shunting inhibition (divisive) with hyperpolarizing inhibition (subtractive). How does each affect the neuron's *f–I* curve?

Q5.14 Why does probabilistic vesicle release produce trial-to-trial variability even when the presynaptic spike train is repeated identically?

Q5.15 In the short-term plasticity model, what parameter regime yields facilitation-dominated behavior? What regime yields depression-dominated behavior?

Q5.16 In vivo neurons often have smaller effective membrane time constants than in vitro. Explain this using the high-conductance-state equation.

6

The Hodgkin–Huxley Model

The leaky integrate-and-fire neuron (Chapter 5) treats the spike as an abstract event. The Hodgkin–Huxley model explains how *the spike is generated: through the coordinated opening and closing of voltage-gated sodium and potassium channels. This chapter develops the gating variable formalism, assembles the full HH equations, and traces the ionic mechanism of the action potential step by step.*

6.1 Voltage-Dependent Conductances

The passive membrane equation (Equation (4.11)) treats conductances as constants. Real neurons have channels whose conductance depends on the membrane potential. When a channel opens or closes in response to voltage, it creates feedback: voltage changes conductance, which changes current, which changes voltage. This feedback is the engine of the action potential.

We write the conductance of a channel population as

$$g(V,t) = \bar{g}\, p_{\text{open}}(V,t) \tag{6.1}$$

where $\bar{g}$ is the maximal conductance (all channels open) and p_{open} is the fraction of channels that are open at voltage V and time t. The modeling problem reduces to specifying how p_{open} depends on voltage and evolves in time.

6.1.1 Persistent versus transient conductances

A *persistent* conductance has a single activation gate: it opens when the membrane depolarizes and stays open as long as the voltage remains high. A *transient* conductance has both activation and inactivation: it opens briefly upon depolarization but then closes again even if the voltage remains elevated. The sodium conductance responsible for the action potential upstroke is transient. The delayed rectifier potassium conductance is persistent (it activates and stays on as long as the membrane is depolarized).

6.1.2 Passive nonlinear background

Not every nonlinear I–V curve comes from voltage-gated channel kinetics. Even a passive membrane can show nonlinear current–voltage behavior through electrodiffusion (GHK current equation) or through asymmetric internal energy barriers (Eyring rate theory), as discussed in Section 4.6. In HH modeling, we usually separate these effects conceptually: passive nonlinearities set baseline membrane transfer properties, while gating dynamics generate the strongly regenerative feedback of spikes.

6.2 Gating Variables

Hodgkin and Huxley modeled each gate as a two-state switch: open or closed. The transition rates between states depend on voltage. For a single gate with opening rate $\alpha(V)$ and closing rate $\beta(V)$, the fraction of open gates x obeys

Gating Variable Kinetics

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \alpha(V)(1-x) - \beta(V)\,x = \frac{x_\infty(V) - x}{\tau_x(V)} \tag{6.2}$$

$x_\infty(V) = \dfrac{\alpha(V)}{\alpha(V)+\beta(V)}$: steady-state activation, $\tau_x(V) = \dfrac{1}{\alpha(V)+\beta(V)}$: time constant.

The two forms of Equation (6.2) are equivalent but emphasize different aspects. The α/β form shows the opening and closing rates explicitly. The x_∞/τ_x form shows that x relaxes exponentially toward its steady-state value $x_\infty(V)$ with time constant $\tau_x(V)$. Both x_∞ and τ_x are functions of voltage that must be measured experimentally.

The steady-state activation $x_\infty(V)$ is typically a sigmoidal function: near zero at hyperpolarized voltages (channels mostly closed) and near one at depolarized voltages (channels mostly open). The time constant $\tau_x(V)$ is peaked—fastest transitions occur at intermediate voltages where both α and β are significant.

6.2.1 Multiple gates per channel

A single channel may have several independent gates that must all be open simultaneously for ions to flow. If a channel has k identical independent gates, the open probability is x^k. Hodgkin and Huxley found that the squid giant axon's potassium channel behaves as if it has

four independent activation gates (n), and the sodium channel has three activation gates (m) and one inactivation gate (h):

$$p_{\text{open}}^{\text{K}} = n^4, \qquad p_{\text{open}}^{\text{Na}} = m^3 h \tag{6.3}$$

The exponents are not arbitrary. The power n^4 produces a sigmoidal onset of potassium current after a voltage step—the current turns on with a delay because all four gates must open. The factorization $m^3 h$ separates fast activation (m rises quickly) from slow inactivation (h falls slowly), which produces the transient character of the sodium current.

6.3 The Hodgkin–Huxley Equations

Assembling the pieces, the Hodgkin–Huxley model consists of the membrane equation with three voltage-dependent conductances—sodium, potassium, and leak:

Hodgkin–Huxley Voltage Equation

$$C_m \frac{\mathrm{d}V}{\mathrm{d}t} = I_{\text{ext}} - \bar{g}_{\text{Na}}\, m^3 h (V - E_{\text{Na}}) - \bar{g}_{\text{K}}\, n^4 (V - E_{\text{K}}) - g_L (V - E_{\text{L}}) \tag{6.4}$$

$\bar{g}_{\text{Na}}, \bar{g}_{\text{K}}$: maximal conductances, g_L: leak conductance, m, h, n: gating variables obeying Equation (6.2).

This is a system of four coupled ordinary differential equations: one for V and one each for m, h, and n. The gating variables evolve according to Equation (6.2) with voltage-dependent rate functions $\alpha_x(V)$ and $\beta_x(V)$ that Hodgkin and Huxley fitted to their voltage clamp data from the squid giant axon.

The model uses parameters measured at 6.3 °C: $C_m = 1\ \mu\text{F/cm}^2$, $\bar{g}_{\text{Na}} = 120\ \text{mS/cm}^2$, $\bar{g}_{\text{K}} = 36\ \text{mS/cm}^2$, $g_L = 0.3\ \text{mS/cm}^2$, $E_{\text{Na}} = +50\ \text{mV}$, $E_{\text{K}} = -77\ \text{mV}$, $E_{\text{L}} = -54.4\ \text{mV}$.

6.4 The Ionic Mechanism of the Action Potential

The beauty of the HH model is that the action potential emerges from the interplay of three gating variables with different speeds. The sequence, once a suprathreshold stimulus depolarizes the membrane, is as follows.

Upstroke. The activation variable m responds to depolarization on a submillisecond timescale. As m rises, m^3 increases rapidly, opening sodium channels. The resulting inward

sodium current depolarizes the membrane further, which opens more sodium channels—a positive feedback loop. The voltage shoots upward toward E_{Na}.

Peak and repolarization. Two slower processes terminate the spike. The inactivation variable h, which was near 1 at rest, begins to fall. As h decreases, sodium channels close even though the membrane is still depolarized. Simultaneously, the potassium activation variable n rises (on a timescale comparable to h), opening delayed rectifier potassium channels. The outward potassium current drives the voltage back down.

Undershoot and recovery. The potassium conductance is still elevated as the voltage drops below rest, producing an afterhyperpolarization. As n slowly returns to its resting value and h slowly recovers from inactivation, the membrane returns to rest and is ready to fire again.

The *refractory period* has a natural explanation: immediately after a spike, h is low (sodium channels inactivated) and n is high (potassium channels still open). A second stimulus during this period fails to trigger a spike because the sodium current is too weak to overcome the elevated potassium current. The absolute refractory period lasts until h has recovered enough to allow any sodium current; the relative refractory period lasts until n has returned to rest and the full sodium current is available.

6.5 Extracting Gating Kinetics from Voltage Clamp Data

Hodgkin and Huxley determined the rate functions $\alpha(V)$ and $\beta(V)$ for each gating variable using a systematic voltage clamp protocol. The procedure, in outline, is:

Step the voltage from a holding potential to a command voltage V_{cmd}. Measure the time course of the resulting current. Use pharmacological blockers (or timing tricks—sodium current is fast, potassium current is slow) to isolate individual ionic components. From the isolated current and the known driving force $(V_{\mathrm{cmd}} - E_{\mathrm{i}})$, extract the conductance $g_i(t) = I_i(t)/(V_{\mathrm{cmd}} - E_{\mathrm{i}})$. The time course of $g_i(t)$ at each voltage step gives $x_\infty(V_{\mathrm{cmd}})$ (the asymptotic value) and $\tau_x(V_{\mathrm{cmd}})$ (the time constant of approach). Repeating at many voltages maps out the full voltage dependence.

For a single gate at fixed command voltage, the solution of Equation (6.2) is

$$x(t) = x_\infty(V_{\mathrm{cmd}}) + (x(0) - x_\infty(V_{\mathrm{cmd}}))e^{-t/\tau_x(V_{\mathrm{cmd}})}. \tag{6.5}$$

This provides a direct fitting template: the asymptote gives x_∞, and the exponential time constant gives τ_x.

6.5.1 Protocol logic: activation versus inactivation

To estimate activation curves, one uses depolarizing *step-up* commands from a hyperpolarized holding potential where activation gates are mostly closed. To estimate inactivation, one uses prepulse protocols: first set a conditioning voltage to place inactivation at a controlled level, then apply a test pulse and measure available current. Step-down protocols are especially useful for recovery from inactivation.

For a persistent current with open probability x^k, the clamped current has the form

$$I(t) = \bar{g}\, x(t)^k\, (V_{\text{cmd}} - E_{\text{rev}}), \tag{6.6}$$

while for transient sodium channels,

$$I_{\text{Na}}(t) = \bar{g}_{\text{Na}}\, m(t)^3 h(t)\, (V_{\text{cmd}} - E_{\text{Na}}), \tag{6.7}$$

so activation and inactivation must be disentangled by protocol design rather than by a single trace alone.

This program turned an abstract channel model into a quantitatively predictive theory. The HH equations, fitted to voltage clamp data, correctly predicted the shape, speed, and threshold of the action potential—a remarkable achievement, done by hand computation before the era of digital computers.

Computationally, this section is the key bridge from experiment to model parameters: it is what makes the HH equations mechanistic rather than phenomenological.

6.6 Beyond Hodgkin–Huxley: The Channel Zoo

The squid giant axon uses only two voltage-gated channel types (Na^+ and delayed rectifier K^+). Real neurons express dozens of channel types, each with different activation thresholds, kinetics, and ionic selectivities. These channels are computational knobs that shape the neuron's firing behavior.

A few important examples: the *A-type potassium current* (I_A) is a transient outward current that activates near threshold. It delays the onset of firing after depolarization, producing a characteristic pause before the first spike. The *M-type potassium current* (I_M) is a slow, persistent outward current that activates during sustained firing and progressively reduces the firing rate—this is *spike-frequency adaptation*. Transient calcium currents (I_{CaT}) can interact with potassium currents to produce *bursting*: clusters of spikes separated by silent intervals.

Additional channel families are equally important. Hyperpolarization-activated mixed-cation current I_h supports subthreshold resonance and post-inhibitory rebound. High-threshold calcium currents can create plateau depolarizations, while calcium-activated potassium currents contribute to burst termination and adaptation.

A useful organizing view is a threshold–timescale map: each current has a characteristic activation voltage and a characteristic kinetic timescale. Fast near-threshold inward currents favor regenerative upstrokes; slower outward currents favor recovery, adaptation, and oscillatory subthreshold dynamics. This map explains why changing channel densities can switch a neuron between integrator-like and resonator-like behavior.

Each added current changes the neuron's dynamical repertoire. Which channels a neuron expresses determines whether it behaves as an integrator or a resonator, whether it adapts or maintains a steady firing rate, and whether it fires single spikes or bursts. We will formalize these distinctions using phase-plane analysis in Chapters 8 to 10.

6.7 Energy Cost of Spikes

Generating an action potential is metabolically expensive. During each spike, sodium ions flow in and potassium ions flow out. The Na^+/K^+-ATPase pump must restore these concentration gradients, consuming ATP in the process.

The energy cost depends on the *overlap* between sodium and potassium currents during the spike. If the sodium current were completely finished before the potassium current began, the charge transfer would be minimal. In reality, there is substantial temporal overlap: sodium is still flowing in while potassium starts flowing out, and this simultaneous flow wastes energy—ions cross the membrane only to be pumped back.

Some neurons are more energy-efficient than others. Neurons with faster sodium inactivation and more precisely timed potassium activation have less current overlap and lower cost per spike. This is not just a curiosity—the brain consumes roughly 20% of the body's energy budget, and a significant fraction goes to ionic pumping. The HH framework lets us quantify this cost by integrating the ionic currents over the course of a spike.

In a conductance-based model, a practical estimate is obtained from the sodium charge transfer:

$$Q_{\mathrm{Na}} = \int_{\mathrm{spike}} I_{\mathrm{Na}}^{\mathrm{in}}(t)\ \mathrm{d}t, \qquad N_{\mathrm{ATP}} \approx \frac{Q_{\mathrm{Na}}}{3e}, \tag{6.8}$$

because one Na/K pump cycle uses one ATP to extrude three sodium ions. Multiplying by the free-energy cost per ATP gives an energy-per-spike estimate.

Empirically, this framework has been used to refine metabolic budgets from early global estimates (Atwell–Laughlin, 2001) to channel-overlap-based efficiency analyses in identified neurons (e.g., Alle–Roth–Geiger, 2009; Carter–Bean, 2009; Sengupta and colleagues, 2010–2013). The main message is consistent: minimizing Na/K current overlap can substantially reduce energy cost without sacrificing fast signaling.

Computationally, energy is not a side constraint; it shapes which dynamical regimes are viable in biological circuits. Channel expression is therefore constrained jointly by coding demands and metabolic cost.

Chapter Summary

Concept	Equation	Key Insight
Gating kinetics	$\dot{x} = \dfrac{x_\infty(V) - x}{\tau_x(V)}$	First-order relaxation, voltage-dependent
HH open probability	$n^4, \quad m^3h$	Multiple independent gates per channel
HH voltage equation	$C_m\dot{V} = I - \bar{g}_{\mathrm{Na}}m^3h(V - E_{\mathrm{Na}}) - \cdots$	Four coupled ODEs generate the spike
AP mechanism	m fast $\rightarrow$ h slow, n slow	Positive then negative feedback
Refractory period	Low h, high n after spike	Sodium inactivated, potassium still open
Clamp extraction	$x(t) = x_\infty + (x_0 - x_\infty)e^{-t/\tau_x}$	Step protocols recover kinetics from data
Metabolic cost	$N_{\mathrm{ATP}} \approx Q_{\mathrm{Na}}/(3e)$	Current overlap sets spike energy efficiency

Conceptual Questions

Q6.1 Explain in your own words why the sodium current is "transient" while the potassium current is "persistent." Which gating variables are responsible for each behavior?

Q6.2 The power n^4 means the potassium conductance turns on with a sigmoid-like delay after a voltage step. Why does raising n to a power produce a delay, and what would the current look like if the power were n^1 instead?

Q6.3 The gating equation can be written in two equivalent forms: the α/β form and the x_∞/τ_x form. What does each form emphasize? Under what circumstances would you prefer one over the other?

Q6.4 Sketch $x_\infty(V)$ and $\tau_x(V)$ for a typical activation gate. Why is x_∞ sigmoidal? Why is τ_x peaked at intermediate voltages?

Q6.5 During the action potential, the membrane voltage briefly exceeds 0 mV but never reaches $E_{\mathrm{Na}} \approx +50$ mV. Why not?

Q6.6 Walk through the action potential mechanism step by step. For each phase (upstroke, peak, repolarization, undershoot, recovery), state which gating variables are changing, in which direction, and how their changes affect the voltage.

Q6.7 Explain the refractory period in terms of the gating variables h and n. Why is the relative refractory period longer than the absolute refractory period?

Q6.8 The HH model is a system of four coupled ODEs. Why does the voltage clamp technique reduce this to three independent first-order equations? How does this simplification make it possible to extract the rate functions $\alpha(V)$ and $\beta(V)$?

Q6.9 Hodgkin and Huxley fitted their model to voltage clamp data at 6.3 °C, then used it to predict the shape and speed of the action potential. Why was this considered such a strong validation of the model?

Q6.10 The sodium current turns on faster than the potassium current. What would happen to the action potential if potassium activated as fast as sodium? What if sodium activated as slowly as potassium?

Q6.11 Adding an A-type potassium current to the HH model delays the first spike after stimulation. Qualitatively, why does a transient outward current near threshold produce this delay?

Q6.12 The M-current produces spike-frequency adaptation. Explain why a slow, persistent outward current causes the interspike interval to lengthen during sustained firing.

Q6.13 The energy cost of a spike depends on the temporal overlap between sodium and potassium currents. Why does greater overlap mean more energy wasted? What would a maximally efficient spike look like in terms of the timing of m, h, and n?

Q6.14 The HH model uses the factorization m^3h for sodium channels, assuming activation and inactivation are independent. What experimental evidence supports this assumption? What would change in the model's predictions if m and h were coupled?

Q6.15 Why do prepulse voltage-clamp protocols isolate inactivation better than a single step from rest? What parameter is being controlled by the prepulse?

Q6.16 Two neurons have similar spike waveforms, but one has much lower Na/K current overlap during spikes. Which neuron is metabolically cheaper per spike, and why?

7

Single Channels and Stochastic Gating

The Hodgkin–Huxley model treats gating variables as continuous fractions: n is the probability that a gate is open, averaged over a large population of channels. But individual channels are discrete—they are either open or closed, and they switch randomly between states. This chapter develops the stochastic description of single-channel gating, shows how the deterministic HH equations emerge as an average over many channels, and introduces conductance-based models that extend HH with additional currents.

7.1 From Deterministic to Stochastic

A voltage clamp experiment on a patch of membrane containing thousands of channels produces smooth, reproducible current traces. But a patch containing a single channel tells a different story: the current jumps abruptly between zero (channel closed) and a fixed value (channel open), at seemingly random times. The channel is a stochastic switch.

The deterministic gating variable $x(t)$ from Equation (6.2) is a population average. It describes the fraction of open gates across many identical channels, not the state of any single channel. When the number of channels is large, the law of large numbers guarantees that the average behaves smoothly. When the number is small—as in a thin dendrite or at a single synapse—the discrete, random switching of individual channels produces significant voltage fluctuations that the deterministic model ignores.

7.2 Stochastic Simulation of a Single Gate

Consider a single gate that can be open (state 1) or closed (state 0). The transition probabilities in a small time step Δt are

$$\Pr(0 \to 1) = \alpha(V)\,\Delta t, \qquad \Pr(1 \to 0) = \beta(V)\,\Delta t \tag{7.1}$$

where $\alpha(V)$ and $\beta(V)$ are the same voltage-dependent rates from the HH model. To simulate the gate, we draw a uniform random number $\eta \in [0, 1]$ at each time step: if the gate is closed and $\eta < \alpha\,\Delta t$, it opens; if the gate is open and $\eta < \beta\,\Delta t$, it closes; otherwise, the state is unchanged.

This is a discrete-time approximation to a continuous-time Markov process. The gate spends random intervals in each state, with mean open time $1/\beta$ and mean closed time $1/\alpha$. The fraction of time spent open, over a long run, converges to $\alpha/(\alpha+\beta) = x_\infty$—exactly the deterministic steady state.

7.3 Many Channels: Recovering the Deterministic Limit

Now suppose the membrane patch contains N identical, independent channels, each with the same gate. At any moment, some number k of the N gates are open, and the macroscopic open fraction is k/N. Each gate switches independently according to Equation (7.1).

As N increases, the fluctuations in k/N shrink—roughly as $1/\sqrt{N}$, by the central limit theorem. For $N \sim 10$, the current trace is noisy. For $N \sim 100$, the ensemble average begins to look smooth. For $N \sim 10{,}000$ (a typical patch), the stochastic fluctuations are negligible and the behavior matches the deterministic HH equations.

This is the connection between microscopic and macroscopic descriptions: the HH gating variable $n(t)$ is the expected value of the open fraction, valid when the channel count is large. When the count is small, you must simulate individual channels.

7.4 The Potassium Channel: From 16 States to 5

The HH potassium channel has four independent gates (n^4). Each gate can be open or closed, giving $2^4 = 16$ possible configurations. But because the gates are identical and independent, only the *number* of open gates matters, not which specific gates are open. We can lump the 16 microstates into 5 macrostates labeled $k = 0, 1, 2, 3, 4$, where k is the number of open gates. The channel conducts only when $k = 4$.

The transition rates between macrostates follow from combinatorics:

$$k \to k+1: \quad (4-k)\,\alpha_n, \qquad k \to k-1: \quad k\,\beta_n \tag{7.2}$$

The factor $(4-k)$ counts the closed gates available to open; the factor k counts the open gates available to close. This 5-state Markov chain is the stochastic counterpart of the deterministic equation $\dot{n} = \alpha_n(1-n) - \beta_n n$.

7.5 The Sodium Channel: Activation and Inactivation

The sodium channel introduces inactivation. In the HH formulation, the open probability is m^3h, where m (activation) and h (inactivation) are assumed independent. This gives

$2^3 \times 2 = 16$ microstates, which can again be lumped based on the number of open activation gates and the state of the inactivation gate.

The independence assumption simplifies the Markov model considerably. In reality, inactivation may depend on the activation state ("state-dependent inactivation"), leading to more complex Markov schemes with additional pathways. Distinguishing between these scenarios requires careful single-channel experiments, often involving analysis of open-time distributions and conditional transition probabilities.

At the macroscopic level, both schemes produce similar current traces—the differences show up primarily in single-channel statistics. For our purposes, the independent m^3h model captures the essential phenomenology: fast activation followed by slower inactivation produces a transient inward current.

7.6 Conductance-Based Models as a Design Toolkit

The HH framework is not limited to two channel types. By adding currents with different kinetics, we can reproduce a wide range of neural firing patterns. The general form is

$$C_m \frac{\mathrm{d}V}{\mathrm{d}t} = I_{\text{ext}} - \sum_i \bar{g}_i \, p_i(V,t)(V - E_i) \tag{7.3}$$

where each current i has its own maximal conductance $\bar{g}_i$, gating function p_i, and reversal potential E_i. Choosing which currents to include—and with what kinetics—determines the neuron's behavior.

7.6.1 The A-current and delayed firing

The *Connor–Stevens model* adds a transient potassium current (I_A) to the standard HH channels. The A-current has a low activation threshold and fast activation, but it also inactivates. When the membrane is depolarized from rest, I_A activates quickly and opposes further depolarization. As the A-current inactivates (over tens of milliseconds), its opposition fades and the neuron can finally fire. The result is a characteristic delay to first spike.

The A-current also shapes the neuron's excitability type. With I_A, the onset of repetitive firing can occur at arbitrarily low frequencies—a hallmark of Type I excitability that we will connect to saddle-node bifurcations in Chapter 10.

7.6.2 Calcium currents and bursting

Adding a transient calcium current (I_{CaT}) introduces a slow inward current that can sustain a plateau depolarization. When this calcium current interacts with a calcium-activated potassium current, the result is *bursting*: the calcium current drives a cluster of rapid spikes, the accumulating potassium current eventually terminates the burst, calcium channels deinactivate during the silent phase, and the cycle repeats.

7.6.3 The M-current and adaptation

The M-type potassium current (I_M) activates slowly during sustained firing. As it builds up, the increasing outward current reduces the membrane's excitability, progressively lengthening interspike intervals. This is spike-frequency adaptation—the firing rate decreases over time even when the input is constant.

Each of these currents acts as a computational knob. The same basic membrane equation, with different channel inventories, can produce integrators, resonators, bursters, or adapting neurons. The dynamical systems analysis of these behaviors is the subject of Chapters 8 to 10.

Chapter Summary

Concept	Equation	Key Insight
Stochastic gate	$\Pr(0\to 1) = \alpha\,\Delta t$	Individual channels switch randomly
Many-channel limit	Fluctuations $\sim 1/\sqrt{N}$	HH is the $N \to \infty$ average
K channel states	$k \to k{+}1$: $(4{-}k)\alpha_n$	16 microstates lumped to 5
General conductance model	$C_m\dot{V} = I - \sum \bar{g}_i p_i (V{-}E_i)$	Channel inventory → firing pattern
A-current	Transient K^+, low threshold	Delayed first spike, Type I excitability

Conceptual Questions

Q7.1 Why does a single-channel recording show discrete jumps between current levels, while a whole-cell recording shows smooth currents? How many channels are needed before the deterministic HH model becomes a good approximation?

Q7.2 The potassium channel has four independent gates, but the Markov model uses only 5 states instead of 16. What assumption allows this simplification, and when might it break down?

Q7.3 Starting from the transition probabilities in Equation (7.1), explain why the long-run fraction of time a gate spends open is $\alpha/(\alpha+\beta)$.

Q7.4 The transition rate from state k to state $k+1$ in the potassium channel Markov model is $(4-k)\alpha_n$. Explain the factor $(4-k)$ in physical terms. Why does the opening rate decrease as more gates are already open?

Q7.5 Fluctuations in the open fraction scale as $1/\sqrt{N}$ for N independent channels. A thin dendrite might have only ~ 10 sodium channels. What consequences would channel noise have for spike initiation in this compartment?

Q7.6 A single-channel recording under voltage clamp shows random open and closed intervals. What information can you extract from the distribution of open times? What about the distribution of closed times?

Q7.7 The HH model assumes that the activation gates (m) and inactivation gate (h) of the sodium channel are independent. In a single-channel Markov model, what would "state-dependent inactivation" look like, and how would it differ from the independent m^3h model?

Q7.8 The general conductance-based model $C_m\dot{V} = I_{\text{ext}} - \sum_i \bar{g}_i\, p_i(V-E_i)$ treats each current as a separate term. What determines how many currents to include in a model? What are the tradeoffs between a minimal model (few currents) and a detailed model (many currents)?

Q7.9 A neuron expressing an A-current shows a long delay before its first spike. Explain qualitatively how a transient outward current near threshold produces this delay, and why the delay disappears for strong stimuli.

Q7.10 How does adding a calcium-activated potassium current produce bursting? Describe the sequence of events during one burst cycle, including the roles of I_{CaT} and $I_{\text{K(Ca)}}$.

Q7.11 Why is the distinction between "macroscopic" (HH-style) and "microscopic" (Markov) descriptions important for modeling synaptic transmission, where the number of channels at a single synapse may be very small?

Q7.12 If you voltage-clamp a membrane patch and average the current over many repeated trials, you recover the deterministic HH prediction. Explain why this ensemble average converges to the deterministic model, and what information is lost in the averaging.

Part III

Dynamical Systems and Excitability

8

Phase-Space Analysis: One Dimension

The Hodgkin–Huxley model has four state variables, making it hard to visualize. Phase-space methods let us understand the qualitative behavior of dynamical systems—stability, threshold, oscillations—from geometry rather than exact solutions. We begin with one-dimensional systems, where the analysis is particularly clean: fixed points, their stability, and how they appear and disappear as parameters change.

8.1 One-Dimensional Flows

A one-dimensional dynamical system has a single state variable x evolving according to

$$\dot{x} = F(x; a) \tag{8.1}$$

where a is a parameter we can control (often the injected current I_{ext}). We cannot always solve this equation in closed form, but we can understand its qualitative behavior by plotting $F(x)$ and reading off the dynamics geometrically.

The key idea is simple: when $F(x) > 0$, x is increasing; when $F(x) < 0$, x is decreasing. We can draw arrows on the x-axis showing the direction of motion. This is the *phase portrait* of the system.

8.2 Fixed Points and Stability

A *fixed point* x^* is a value where $F(x^*) = 0$—the system is in equilibrium. What happens if we perturb slightly away from x^*? We linearize: set $x = x^* + \delta x$ and expand to first order:

$$\dot{\delta x} \approx F'(x^*)\,\delta x \tag{8.2}$$

The perturbation grows or decays exponentially with rate $F'(x^*)$. If $F'(x^*) < 0$, perturbations decay and the fixed point is *stable*. If $F'(x^*) > 0$, perturbations grow and the fixed point is *unstable*.

Graphically, this has a clean interpretation. At a stable fixed point, F crosses zero from above to below (the curve goes from positive to negative as x increases). At an unstable fixed point, F crosses from below to above. You can read off stability directly from the plot of $F(x)$ without computing any derivatives.

8.2.1 The potential picture

When $F(x) = -\mathrm{d}U/\mathrm{d}x$ for some function $U(x)$, the dynamics become $\dot{x} = -U'(x)$—the system rolls downhill on the "energy landscape" $U(x)$. Stable fixed points are minima of U; unstable fixed points are maxima. A ball placed on a hilltop (unstable) rolls down to the nearest valley (stable).

Not every system admits a potential, but the picture is a powerful intuition even when it does not literally apply. In one dimension, the qualitative behavior is always potential-like: trajectories flow toward stable fixed points and away from unstable ones, and no oscillations are possible.

8.3 Threshold as an Unstable Fixed Point

We can now give "threshold" a precise meaning. Consider a simplified neuron model with a single voltage variable:

$$C_m \dot{V} = -g_L(V - E_\mathrm{L}) - \bar{g}_\mathrm{Na}\, m_\infty(V)\,(V - E_\mathrm{Na}) + I_\mathrm{ext} \tag{8.3}$$

This is a persistent sodium model: the sodium activation m is so fast that we replace it by its steady-state value $m_\infty(V)$, while the slower variables (h, n) are frozen at their resting values. The result is a single ODE in V.

The function $F(V)$ on the right-hand side is the net current. The leak current is stabilizing (roughly linear, pulling V toward E_L). The persistent sodium current is regenerative: as V increases, $m_\infty(V)$ increases, which further increases the inward current—a positive feedback loop.

For appropriate parameters, $F(V) = 0$ has three solutions: a stable resting state, an unstable equilibrium, and a regime where the voltage runs away toward E_Na. The unstable equilibrium is the *threshold*: perturbations below it decay back to rest; perturbations above it trigger a spike. Threshold is not a magic number—it is the voltage at which the sodium positive feedback overwhelms the leak.

8.4 Saddle-Node Bifurcation

As the parameter a (say, the injected current) changes, fixed points can move, appear, or disappear. A *saddle-node bifurcation* occurs when a stable and an unstable fixed point collide and annihilate each other.

Saddle-Node Bifurcation Conditions

At a saddle-node bifurcation (x_f, a_{SN}):

$$F(x_f) = 0, \qquad \left.\frac{\partial F}{\partial x}\right|_{x_f} = 0, \qquad \left.\frac{\partial^2 F}{\partial x^2}\right|_{x_f} \neq 0, \qquad \left.\frac{\partial F}{\partial a}\right|_{a_{\mathrm{SN}}} \neq 0 \tag{8.4}$$

The first two conditions say the fixed point is non-hyperbolic (the linearization has a zero eigenvalue). The last two are non-degeneracy conditions ensuring a genuine bifurcation.

Just below the bifurcation (say, for $a < a_{\mathrm{SN}}$), two fixed points exist: rest and threshold. At $a = a_{\mathrm{SN}}$, they merge into a single half-stable point. For $a > a_{\mathrm{SN}}$, both fixed points are gone, and there is no stable rest—the system must do something else (typically, fire repetitively).

8.4.1 The ghost of the fixed point

Just above the bifurcation, the fixed points no longer exist, but the function $F(x)$ is still close to zero in the region where they used to be. Trajectories pass through this region very slowly—they linger near the "ghost" of the annihilated fixed point. This produces a long latency before the first spike: the neuron sits near the former resting state for hundreds of milliseconds before finally escaping. The closer the current is to the bifurcation value, the longer the latency. This ghost effect is a distinctive signature of saddle-node bifurcations in neurons.

Chapter Summary

Concept	Equation	Key Insight
1D flow	$\dot{x} = F(x; a)$	Direction of motion from sign of F
Stability	$F'(x^*) < 0$: stable	Slope at zero crossing determines stability
Potential picture	$\dot{x} = -U'(x)$	Stable = valley, unstable = hilltop
Threshold	Unstable fixed point	Separates basins of rest and spike
Saddle-node	$F = 0, F' = 0$	Stable and unstable FPs collide and vanish
Ghost effect	Slow passage near former FP	Long latency near bifurcation

Conceptual Questions

Q8.1 In a one-dimensional system, can you have oscillations? Why or why not? What is the minimum dimension needed for sustained oscillations?

Q8.2 Given a plot of $F(x)$ versus x, how can you determine the location and stability of all fixed points by inspection? What graphical feature tells you whether a fixed point is stable or unstable?

Q8.3 The linearization near a fixed point gives $\dot{\delta x} \approx F'(x^*)\,\delta x$. What does it mean for $F'(x^*) = 0$? Why is this case "non-hyperbolic," and why can't linearization tell you the stability?

Q8.4 Explain in your own words why threshold in the persistent sodium model is an unstable fixed point. What two competing forces determine where this point sits?

Q8.5 In the persistent sodium model, sketch $F(V)$ and identify the three fixed points. What happens to the shape of $F(V)$ as you increase I_{ext}? At what current do two of the fixed points disappear?

Q8.6 The potential picture interprets stable fixed points as valleys and unstable ones as hilltops. In this picture, what does a saddle-node bifurcation look like? Sketch the

energy landscape $U(x)$ for three values of the parameter: below, at, and above the bifurcation.

Q8.7 Near a saddle-node bifurcation, the neuron shows a long latency before the first spike. Where does this delay come from, and how does it change as the injected current moves further above the bifurcation value?

Q8.8 A saddle-node bifurcation requires four conditions (Equation (8.4)). Explain in physical terms what each condition means. Why are the non-degeneracy conditions ($F'' \neq 0$, $\partial F/\partial a \neq 0$) necessary?

Q8.9 Compare the threshold concept in the LIF model (a fixed voltage V_{th}) with the threshold concept in the persistent sodium model (an unstable fixed point). Which is more realistic, and why?

Q8.10 Not every one-dimensional system has a potential function $U(x)$. Why not? Give an example of a one-dimensional system $\dot{x} = F(x)$ where a potential always exists, and explain why this makes oscillations impossible.

Q8.11 A neuron model has a single stable fixed point at rest and no other fixed points. Can it produce a spike in response to a brief stimulus? Why or why not?

9

Phase-Space Analysis: Two Dimensions

One-dimensional systems cannot oscillate. To understand repetitive firing, we need at least two dimensions—typically voltage and a recovery variable. This chapter develops the tools of two-dimensional phase-plane analysis: vector fields, nullclines, fixed point classification via the Jacobian, and limit cycles.

9.1 Two-Dimensional Systems

A two-dimensional dynamical system has the form

$$\dot{x} = f(x, y), \qquad \dot{y} = g(x, y) \tag{9.1}$$

The state of the system is a point (x, y) in the *phase plane*. At each point, the functions f and g define a velocity vector $(\dot{x}, \dot{y})$—the direction and speed the state moves. The collection of all these vectors is the *vector field*. A trajectory is a curve that follows the vector field from some initial condition.

For neuron models, the two variables are typically V (membrane voltage) and a recovery variable—often a potassium gating variable n or a generic slow variable w that provides negative feedback. The voltage is fast (it changes quickly); the recovery variable is slow (it changes on a longer timescale). This separation of timescales is what makes neuroscience models amenable to phase-plane analysis.

9.2 Nullclines

A *nullcline* is a curve in the phase plane where one of the derivatives vanishes:

$$x\text{-nullcline:} \quad f(x, y) = 0, \qquad y\text{-nullcline:} \quad g(x, y) = 0 \tag{9.2}$$

On the x-nullcline, $\dot{x} = 0$, so the vector field is purely vertical (the state moves only in the y direction). On the y-nullcline, $\dot{y} = 0$, so the vector field is purely horizontal. Where the two nullclines intersect, both derivatives are zero: these are the *fixed points* of the system.

Nullclines are the skeleton of the phase portrait. Once you draw them, you can determine the direction of the vector field in each region by checking the signs of f and g. This lets you sketch the qualitative flow without solving a single equation.

For a neuron model with voltage V and recovery variable n, the V-nullcline is often a cubic-shaped curve (reflecting the N-shape of the total current as a function of voltage), while the n-nullcline is a monotonically increasing function (higher voltage means larger steady-state activation). Their intersections determine how many fixed points exist and how the system behaves.

9.3 Fixed Point Classification

Near a fixed point (x^*, y^*), we linearize by setting $(x, y) = (x^*, y^*) + (\delta x, \delta y)$:

$$\begin{pmatrix} \dot{\delta x} \\ \dot{\delta y} \end{pmatrix} \approx J \begin{pmatrix} \delta x \\ \delta y \end{pmatrix}, \qquad J = \begin{pmatrix} \partial f/\partial x & \partial f/\partial y \\ \partial g/\partial x & \partial g/\partial y \end{pmatrix}_{(x^*, y^*)} \tag{9.3}$$

The eigenvalues of the Jacobian J determine the local geometry. The characteristic equation is $\lambda^2 - \tau\lambda + \Delta = 0$, where $\tau = \mathrm{tr}(J)$ and $\Delta = \det(J)$.

Trace–Determinant Classification

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2} \tag{9.4}$$

$\tau = \mathrm{tr}(J)$: controls stability (negative → stable). $\Delta = \det(J)$: sign distinguishes saddles from nodes/spirals.

$\tau^2 - 4\Delta$: sign distinguishes nodes (real λ) from spirals (complex λ).

The classification is as follows. If $\Delta < 0$, the eigenvalues are real with opposite signs: the fixed point is a *saddle*. If $\Delta > 0$ and $\tau < 0$, both eigenvalues have negative real parts: the fixed point is a *stable node* (real eigenvalues) or *stable spiral* (complex eigenvalues, meaning damped oscillations). If $\Delta > 0$ and $\tau > 0$, the fixed point is an unstable node or spiral. The boundary $\tau^2 = 4\Delta$ separates nodes from spirals.

9.3.1 Why saddles matter for neurons

Saddle points appear frequently in neuron models, and they play a disproportionately important role. A saddle has one attracting direction and one repelling direction. The *stable manifold* of a saddle—the curve along which trajectories approach it—acts as a separatrix, dividing the phase plane into qualitatively different regions. In neuron models, this separatrix is the

geometric object behind "threshold": trajectories on one side return to rest, trajectories on the other side produce a spike.

This is a much richer picture of threshold than the one-dimensional version. In two dimensions, threshold is not a single voltage value—it is a *curve* in the (V, n) plane. A stimulus that brings the state across this curve triggers a spike; one that does not, no matter how close, does not.

9.4 Limit Cycles

A *limit cycle* is a closed trajectory in the phase plane: the state returns to its starting point after a period T, and the motion repeats indefinitely. A *stable* limit cycle attracts nearby trajectories: if the state starts near the cycle, it spirals onto it. A stable limit cycle is the mathematical object behind repetitive spiking—each orbit around the cycle corresponds to one action potential.

Limit cycles cannot exist in one dimension (trajectories on a line cannot return to their starting point without stopping at a fixed point in between). They are an essentially two-dimensional phenomenon.

As a parameter (say, the injected current) increases, a stable fixed point can lose stability and give way to a limit cycle. The specific way this transition happens—the *bifurcation*—determines the neuron's excitability type, which is the subject of Chapter 10.

9.5 A Worked Example: The Persistent Sodium Model in 2D

To make these ideas concrete, consider a two-variable reduction of the Hodgkin–Huxley model with persistent sodium and potassium:

$$C_m \dot{V} = -\bar{g}_{\mathrm{Na}}\, m_\infty(V)\,(V - E_{\mathrm{Na}}) - \bar{g}_{\mathrm{K}}\, n\,(V - E_{\mathrm{K}}) - g_L(V - E_{\mathrm{L}}) + I_{\mathrm{ext}} \tag{9.5}$$

$$\dot{n} = \frac{n_\infty(V) - n}{\tau_n(V)} \tag{9.6}$$

The V-nullcline ($\dot{V} = 0$) is the curve $n = n_V(V)$ obtained by solving the right-hand side of Equation (9.5) for n. It is typically cubic-shaped: an N-curve with a left branch (rest), a middle branch (unstable), and a right branch (excited). The n-nullcline ($\dot{n} = 0$) is $n = n_\infty(V)$, which is a monotone sigmoid.

Their intersection gives the fixed points. When the current I_{ext} is small, the intersection is on the left branch—a stable resting state. As current increases, the V-nullcline shifts (current pushes the curve up), and the fixed point moves along the nullcline. Beyond a critical current, the fixed point can lose stability and a limit cycle appears—the neuron fires repetitively.

The full action potential trajectory traces a large loop in the (V, n) plane: rapid depolarization (fast V increase with n lagging behind), followed by repolarization (increasing n opens potassium channels), afterhyperpolarization (high n, low V), and recovery (n slowly decreases back to baseline). The nullclines organize this loop.

Chapter Summary

Concept	Equation	Key Insight
2D system	$\dot{x} = f(x, y), \dot{y} = g(x, y)$	Trajectories in phase plane
Nullclines	$f = 0, g = 0$	Skeleton of the flow; intersections = FPs
Jacobian	$J = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}$	Eigenvalues classify local dynamics
Trace–det	$\Delta < 0$: saddle; $\tau < 0$: stable	Quick classification without computing λ
Limit cycle	Closed orbit, period T	Repetitive spiking = stable limit cycle

Conceptual Questions

Q9.1 Why can oscillations exist in two dimensions but not in one? What geometric property of trajectories in 2D makes this possible?

Q9.2 You are given the nullclines of a 2D neuron model. Without computing anything, how can you determine the direction of the vector field in each region of the phase plane?

Q9.3 Explain why nullcline intersections are fixed points. If two nullclines intersect three times, what can you say about the stability of the three fixed points?

Q9.4 The Jacobian matrix J evaluated at a fixed point has trace τ and determinant Δ. Explain the physical meaning of each: what does τ control, and what does Δ control?

Q9.5 A fixed point has complex eigenvalues with negative real part. Describe the trajectory of a perturbation near this fixed point. How does the behavior change if the real part becomes positive?

Q9.6 A saddle point's stable manifold acts as a separatrix in neuron models. Why is this a more accurate description of "threshold" than a single voltage value? How does the separatrix relate to the concept of threshold in the one-dimensional analysis of Chapter 8?

Q9.7 In the persistent sodium model, the V-nullcline is cubic-shaped. What happens to the fixed point as I_{ext} increases? Describe the sequence of qualitative changes.

Q9.8 A limit cycle is a closed orbit in the phase plane. What is the difference between a stable and an unstable limit cycle? How do nearby trajectories behave in each case?

Q9.9 The V-nullcline of a typical neuron model has a cubic (N-shaped) form. Explain why: what biophysical features of the voltage equation produce the left branch, the middle branch, and the right branch?

Q9.10 In the 2D persistent sodium model, trace the action potential trajectory in the (V, n) plane. Why does the trajectory form a large loop rather than returning along the same path? What role does the separation of timescales play?

Q9.11 Two neurons can have identical nullclines but different timescale separations (one has τ_n much larger than the other). How does the timescale ratio affect the shape of the action potential in the phase plane?

Q9.12 Explain why increasing I_{ext} shifts the V-nullcline but not the n-nullcline. What does this tell you about how external input changes the number and location of fixed points?

Q9.13 Can a 2D system have more than one stable limit cycle? Can a stable limit cycle and a stable fixed point coexist? What does this imply for the neuron's response to stimuli of different strengths?

10

Excitability Types: Integrators and Resonators

Not all neurons start firing the same way. Some can fire at arbitrarily low rates, while others jump abruptly to a minimum nonzero frequency. This distinction traces back to the type of bifurcation that creates the limit cycle: a saddle-node on an invariant circle produces Type I (integrator) behavior, while a Hopf bifurcation produces Type II (resonator) behavior. These two classes differ in everything from threshold geometry to coding strategy.

10.1 Two Roads to Repetitive Firing

As the injected current increases, a neuron transitions from silence to repetitive firing. The transition occurs when the stable resting state loses stability and a limit cycle appears. But there are fundamentally different ways this can happen, and the mechanism determines the neuron's computational character.

The two main bifurcation types are the *saddle-node on an invariant circle* (SNIC) and the *Hopf bifurcation*. Each creates a limit cycle, but with profoundly different properties.

10.2 Type I: Saddle-Node on Invariant Circle

In a SNIC bifurcation, the resting state (a stable node) and a nearby saddle point sit on a closed curve in the phase plane. As the current increases, the node and saddle approach each other, collide, and annihilate—exactly as in the one-dimensional saddle-node (Section 8.4), except that now the annihilation happens on a circle. Once both fixed points are gone, trajectories travel around the entire circle: a limit cycle is born.

The distinctive feature of the SNIC is that the limit cycle's period is *infinite* at onset. Just above the bifurcation, trajectories still pass slowly through the ghost of the annihilated fixed points, spending a long time in the bottleneck region. The resulting firing rate can be made arbitrarily small by bringing the current just above threshold. This gives a continuous, smooth *f*–*I* curve that starts from zero.

10.2.1 Properties of Type I neurons

Type I neurons behave as *integrators*: they accumulate input over time and fire when enough has been collected. Their key properties are:

The *f–I* curve is continuous at onset—the firing rate starts from zero and increases smoothly with current. There is a well-defined threshold, corresponding to the separatrix (the saddle's stable manifold) that exists just below the bifurcation. Subthreshold responses are mostly monotone rather than oscillatory: excitation pushes the state toward threshold and inhibition pulls it back. The neuron encodes stimulus intensity through its firing *rate*: stronger input produces higher rates, with no preferred frequency.

Near threshold, the ghost of the saddle-node produces a long latency to the first spike—the neuron sits near rest for hundreds of milliseconds before firing. This latency decreases sharply as the current is increased.

10.3 Type II: Hopf Bifurcation

In a Hopf bifurcation, the resting state (a stable spiral) loses stability as a pair of complex conjugate eigenvalues crosses the imaginary axis. The real part of the eigenvalues changes sign from negative to positive, and the stable spiral becomes an unstable spiral. Simultaneously, a limit cycle appears around the newly unstable fixed point.

In a *supercritical* Hopf bifurcation, the limit cycle is stable and has small amplitude at birth, growing as the parameter moves further past the bifurcation. In a *subcritical* Hopf, the limit cycle is unstable (it acts as a separatrix), and the neuron jumps abruptly to a large-amplitude oscillation. Either way, the limit cycle has a *nonzero frequency* at onset, set by the imaginary part of the eigenvalues at the bifurcation.

10.3.1 Properties of Type II neurons

Type II neurons behave as *resonators*: they have a preferred frequency and respond best to inputs at that frequency. Their key properties are:

The *f–I* curve has a discontinuity at onset—the firing rate jumps from zero to a minimum nonzero value. There is no clean threshold; subthreshold oscillations exist below the bifurcation (the stable spiral produces damped oscillations that can be excited by brief inputs). Inputs at the neuron's resonant frequency are preferentially amplified, while inputs at other frequencies are suppressed. The neuron encodes stimulus timing through phase-locking to periodic inputs.

Post-inhibitory rebound is a hallmark of resonators: a brief hyperpolarizing pulse can

trigger a spike after the pulse ends, because the release from inhibition interacts with the subthreshold oscillation to push the voltage above threshold.

Close to Hopf onset, responses can be *graded* rather than all-or-none: different perturbation amplitudes can produce different spike amplitudes or spike failures before the trajectory settles onto the limit cycle. Pulse timing also matters strongly: inputs that align with the intrinsic oscillation phase can trigger spikes efficiently, while off-phase pulses may be ineffective.

10.4 Global Orbit Geometry Near Threshold

Local eigenvalue analysis explains stability, but global phase-plane structures determine threshold geometry and spike initiation pathways. Two important objects are heteroclinic and homoclinic orbits.

A *heteroclinic orbit* connects two different equilibria (typically along stable/unstable manifolds). A *homoclinic orbit* leaves and returns to the same saddle equilibrium. In excitability models, these orbits organize transitions between rest, spike initiation, and large-amplitude oscillations.

Near SNIC regimes, the saddle's stable manifold forms the threshold manifold and its unstable manifold helps shape trajectories toward the spiking branch. Near Hopf regimes, threshold becomes less geometric and more phase dependent, because trajectories spiral and threshold cannot be summarized by a single voltage or a single separatrix curve.

Computationally, these global structures explain why two models with similar local linearization can still have very different spike-triggering behavior.

10.5 Reading Bifurcation Diagrams

A bifurcation diagram plots equilibria and periodic orbits as a control parameter (often I_{ext}) varies. Solid branches denote stable solutions; dashed branches denote unstable ones. For Type I onset (SNIC), the oscillation frequency approaches zero at threshold. For Type II onset (Hopf), oscillations emerge at nonzero frequency.

In practice, three visual checks are especially useful:

1. Track where the resting fixed point loses stability.
2. Check whether oscillation amplitude appears continuously (supercritical Hopf) or by jump (subcritical Hopf / global transition).
3. Check whether onset frequency tends to 0 (SNIC-like) or stays finite (Hopf-like).

These diagnostics are what connect numerical continuation plots to experimentally measurable *f–I* curves, latency, and resonance.

10.6 Integrator versus Resonator: A Comparison

The distinction between integrators and resonators is not merely a classification exercise—it reflects fundamentally different computational strategies.

Property	**Integrator (Type I)**	**Resonator (Type II)**
Bifurcation	SNIC	Hopf
f–I onset	Continuous from 0 Hz	Jump to nonzero frequency
Threshold	Well-defined separatrix	No sharp boundary
Subthreshold dynamics	Monotone decay	Damped oscillations
Preferred frequency	None (broadband)	Yes (resonance)
Coding strategy	Rate coding	Temporal / coincidence coding
Post-inhibitory rebound	Absent	Present
Latency near threshold	Long (ghost effect)	Short

The standard Hodgkin–Huxley model, with its original squid axon parameters at 6.3 °C, exhibits Type II behavior: the onset of firing involves a Hopf-like bifurcation, and the *f–I* curve jumps to a nonzero frequency. However, varying the channel densities or temperature can shift the model to Type I behavior. Biology is not locked into one class—different neurons in the brain use different strategies depending on their computational role.

10.6.1 Threshold manifold versus threshold voltage

For integrators close to SNIC, threshold is well approximated by a manifold (the stable manifold of a saddle): crossing it predicts spike initiation robustly. For resonators close to Hopf, this manifold picture weakens; threshold depends strongly on state-space phase and input timing. This is why "threshold voltage" is often variable in resonator-like neurons.

Computationally, this distinction clarifies why some neurons are well described by scalar threshold models, while others require at least two-dimensional state tracking.

10.7 Reduced Models

Full conductance-based models have many parameters and are hard to analyze. Reduced models capture the essential dynamics in two variables, making phase-plane analysis tractable.

10.7.1 The Izhikevich model

The Izhikevich model uses a quadratic voltage equation with a linear recovery variable:

$$C_m \dot{v} = k(v - v_r)(v - v_t) - u + I \tag{10.1}$$

$$\dot{u} = a[b(v - v_r) - u] \tag{10.2}$$

with reset: if $v \geq v_{\text{peak}}$, set $v \leftarrow c$ and $u \leftarrow u + d$. The parameters a, b, c, d control the timescale of recovery, the subthreshold behavior, the reset voltage, and the post-spike adaptation. By tuning these four parameters, the model can reproduce both Type I and Type II excitability, bursting, adaptation, and many other firing patterns observed in real neurons.

The model bridges the gap between the biophysical detail of Hodgkin–Huxley and the simplicity of integrate-and-fire. The quadratic nonlinearity in v captures the essential positive feedback of sodium activation without requiring explicit gating variables.

10.7.2 The theta model

The *theta model* places the dynamics on a circle:

$$\dot{\theta} = (1 - \cos\theta) + (1 + \cos\theta)\, I \tag{10.3}$$

A "spike" corresponds to θ passing through π. The circle topology means there is no need for an artificial reset rule—the system wraps around naturally. The theta model is the canonical normal form for the SNIC bifurcation. Below threshold $(I < 0)$, there are two fixed points on the circle; above threshold $(I > 0)$, they annihilate and the phase θ increases monotonically, producing periodic spiking.

10.7.3 The resonate-and-fire model

A complementary canonical model for resonators is the *resonate-and-fire* model. In complex notation:

$$\dot{z} = (\lambda + i\omega_0)z + I(t), \tag{10.4}$$

with a spike emitted when
$Re(z)$ crosses threshold, followed by reset. The parameter

omega$_0$ sets intrinsic subthreshold oscillation frequency, and
lambda < 0 sets damping.

Unlike LIF-like integrators, this model explicitly captures phase-sensitive excitability, rebound, and frequency-selective response. It is therefore a convenient reduced model for Type II-like dynamics when full conductance models are too costly.

Chapter Summary

Concept	Equation / Property	Key Insight
SNIC bifurcation	Node + saddle annihilate on circle	Type I: continuous *f*–*I*, arbitrarily low rate
Hopf bifurcation	Complex λ cross imaginary axis	Type II: nonzero onset frequency, resonance
Integrator	Rate coding, clear threshold	Accumulates input; fires when enough
Resonator	Temporal coding, damped oscillations	Responds best at preferred frequency
Global orbits	Heteroclinic / homoclinic geometry	Organize threshold and trajectory structure
Bifurcation diagram	Stable/unstable branches vs I	Links phase-plane theory to measurable *f*–*I*
Izhikevich model	$C_m\dot{v} = k(v - v_r)(v - v_t) - u + I$	Four parameters reproduce many patterns
Theta model	$\dot{\theta} = (1 - \cos\theta) + (1 + \cos\theta)I$	SNIC normal form; no reset needed
Resonate-and-fire	$\dot{z} = (\lambda + i\omega_0)z + I$	Canonical phase-sensitive resonator model

Conceptual Questions

Q10.1 Why does the SNIC bifurcation produce a continuous *f*–*I* curve starting from zero, while the Hopf bifurcation produces a jump to a nonzero frequency? Connect your answer to the dynamics near the bifurcation.

Q10.2 In a SNIC bifurcation, the period of the limit cycle is infinite at onset. Where does the trajectory spend most of its time near the bifurcation, and why?

Q10.3 Explain the difference between a supercritical and a subcritical Hopf bifurcation. Which one produces a smooth transition to small-amplitude oscillations? Which produces an abrupt jump to large-amplitude spiking?

Q10.4 A resonator neuron shows post-inhibitory rebound: a brief hyperpolarizing pulse can trigger a spike. Why does this happen, and why is it absent in integrators?

Q10.5 An integrator neuron uses rate coding: stimulus intensity is encoded in firing rate. A resonator uses temporal coding: stimulus timing is encoded in spike timing relative to the input. Explain why each coding strategy follows naturally from the underlying bifurcation.

Q10.6 Two excitatory inputs arrive at an integrator neuron, one 5 ms after the other. Compare the neuron's response to the case where both inputs arrive simultaneously. Now repeat the comparison for a resonator neuron. How does the integration window differ?

Q10.7 The comparison table lists "preferred frequency: none" for integrators and "preferred frequency: yes" for resonators. What physical feature of the subthreshold dynamics gives rise to a preferred frequency? Where does this frequency come from in terms of the eigenvalues?

Q10.8 The standard Hodgkin–Huxley model is Type II, but varying channel densities can make it Type I. What does this tell you about the relationship between channel biophysics and excitability type?

Q10.9 The Izhikevich model uses a quadratic nonlinearity $k(v - v_r)(v - v_t)$ instead of explicit gating variables. What biophysical process does this term approximate, and why is a quadratic sufficient?

Q10.10 The theta model $\dot{\theta} = (1 - \cos\theta) + (1 + \cos\theta)I$ lives on a circle. Why does the circle topology eliminate the need for an explicit reset rule? What are the fixed points when $I < 0$, and what happens to them when $I > 0$?

Q10.11 What is the practical difference between saying "threshold is a manifold" and "threshold is phase dependent"? Give one implication for experiments with brief pulses.

Q10.12 In a bifurcation diagram, how would you distinguish a SNIC-like onset from a Hopf-like onset using frequency behavior near threshold?

Q10.13 A neuron in the brain might switch between integrator and resonator behavior depending on neuromodulatory state. What biophysical change could produce this switch? Give an example involving a specific channel type.

Q10.14 The LIF model from Chapter 5 has a continuous f–I curve starting from zero. Based on this, is the LIF model Type I or Type II? What bifurcation does the threshold-and-reset rule implicitly implement?

Part IV

Spatial and Network Models

11

Cable Theory: Derivation

So far we have treated neurons as single electrical compartments—point neurons with a uniform voltage. Real neurons are spatially extended: a synaptic input at a distal dendrite must travel hundreds of micrometers to reach the soma, losing amplitude and gaining delay along the way. Cable theory provides the mathematical framework for understanding how voltage varies in both space and time along a neurite.

11.1 Why Cable Theory?

The point-neuron model assumes that every part of the cell is at the same voltage. For a compact soma, this is reasonable. But dendrites can extend over a millimeter, and their thin geometry means significant axial resistance. A synaptic potential generated at a distal dendritic spine is attenuated and delayed by the time it reaches the soma. The same physics governs action potential propagation along axons.

Cable theory answers two fundamental questions. First, how does a synaptic potential spread passively from its origin to the soma? Second, how does an action potential propagate along an axon? Both are problems of voltage varying in space and time, and both require treating the neurite as a distributed electrical structure.

11.2 The Cable as a Distributed Circuit

We model a neurite (dendrite or axon) as a cylinder of radius a and length much greater than a. The cytoplasm inside has resistivity r_L (typically $\sim 1\ \Omega\,\mathrm{m}$). The membrane has a specific capacitance c_m and a specific resistance r_m. We treat the extracellular space as having negligible resistance (a good approximation for most situations).

Consider a short segment of length Δx. Four currents act on it: the *axial current* $I_a(x)$ flowing in from the left, the axial current $I_a(x + \Delta x)$ flowing out to the right, the *membrane current* I_m leaking through the membrane, and any *injected current* I_e from an electrode.

The axial current obeys Ohm's law. The resistance of a cylinder of length Δx and cross-

section πa^2 is $r_L \, \Delta x/(\pi a^2)$, so

$$I_a(x) = -\frac{\pi a^2}{r_L}\frac{\partial V}{\partial x} \tag{11.1}$$

The capacitive current charging the membrane of the segment is $c_m \cdot 2\pi a \, \Delta x \cdot \partial V/\partial t$. The membrane current through ion channels is $i_m \cdot 2\pi a \, \Delta x$, where i_m is the membrane current density.

Applying Kirchhoff's current law—current in must equal current out plus current stored—and taking the limit $\Delta x \to 0$:

The Cable Equation

$$c_m \frac{\partial V}{\partial t} = \frac{a}{2r_L}\frac{\partial^2 V}{\partial x^2} - i_m + i_e \tag{11.2}$$

c_m: specific membrane capacitance (F/cm^2), r_L: axial resistivity (Ω cm), a: fiber radius, i_m: membrane current density, i_e: injected current density.

The new ingredient compared to the point-neuron membrane equation is the spatial diffusion term $\frac{a}{2r_L}\frac{\partial^2 V}{\partial x^2}$. This term describes axial current flow: voltage differences between neighboring segments drive current along the cylinder. The cable equation is a partial differential equation in space and time—fundamentally harder than the ordinary differential equations of point-neuron models.

11.3 The Passive Cable: Length and Time Constants

For a passive (linear) membrane, the membrane current is simply $i_m = (V - V_{\text{rest}})/r_m$. Writing $v = V - V_{\text{rest}}$ for the deviation from rest, the cable equation becomes

Passive Cable Equation

$$\tau_m \frac{\partial v}{\partial t} = \lambda^2 \frac{\partial^2 v}{\partial x^2} - v + r_m \, i_e \tag{11.3}$$

$\tau_m = r_m c_m$: membrane time constant, $\lambda = \sqrt{\frac{r_m \, a}{2 \, r_L}}$: length constant (space constant).

Two constants govern the behavior. The *membrane time constant* $\tau_m = r_m c_m$ sets the temporal scale—how quickly voltage changes at any point. The *length constant* λ sets the spatial scale—how far voltage spreads before decaying significantly.

The length constant has a clear physical interpretation: it is the distance at which steady-state voltage decays to $1/e \approx 37\%$ of its value at the injection site. Typical values are $\lambda \sim$ 0.1–1 mm. The length constant increases with fiber radius (larger cross-section means lower axial resistance) and with membrane resistance (less leakage means voltage spreads further). This is why thick axons conduct better than thin ones.

11.4 Steady-State Solution: Infinite Cable

For a constant current injected at $x = 0$ into an infinitely long cable, the steady-state ($\partial v/\partial t = 0$) solution is

$$v(x) = \frac{I_e R_\lambda}{2} e^{-|x|/\lambda} \tag{11.4}$$

where R_λ is the input resistance of a semi-infinite cable. The voltage decays exponentially in both directions from the injection site with space constant λ. The factor of 2 in the denominator reflects that current flows in both directions.

This exponential decay is the hallmark of passive cable conduction. A synapse one length constant away from the soma delivers only 37% of its voltage to the soma. A synapse two length constants away delivers only 14%. This is why synaptic location matters: proximal synapses have much stronger influence on the soma than distal ones, all else being equal.

11.5 Finite Cable with Sealed End

Real dendrites are not infinite—they terminate. A common boundary condition is the *sealed end*: no current flows out of the tip ($\partial v/\partial x = 0$ at $x = L$). For a finite cable of length L with current injected at $x = 0$ and a sealed end at $x = L$, the steady-state solution involves hyperbolic functions:

$$v(x) = v(0) \frac{\cosh((L - x)/\lambda)}{\cosh(L/\lambda)} \tag{11.5}$$

When $L \gg \lambda$, this approaches the infinite-cable exponential. When $L \lesssim \lambda$, the sealed end reflects current back, and the voltage at the tip is nearly as large as at the injection site—the entire cable is "electrically compact." The ratio L/λ is the *electrotonic length*: the cable's physical length measured in units of the space constant. A dendrite with $L/\lambda < 0.5$ behaves nearly like a point neuron; one with $L/\lambda > 2$ shows significant spatial filtering.

Chapter Summary

Concept	Equation	Key Insight
Cable equation	$c_m \frac{\partial V}{\partial t} = \frac{a}{2r_L}\frac{\partial^2 V}{\partial x^2} - i_m + i_e$	Voltage varies in space and time
Length constant	$\lambda = \sqrt{\frac{r_m a}{2r_L}}$	Spatial decay scale; $\propto \sqrt{a}$
Time constant	$\tau_m = r_m c_m$	Temporal decay scale
Infinite cable	$v(x) \propto e^{-\lvert x\rvert/\lambda}$	Exponential spatial attenuation
Sealed end	$v(x) \propto \cosh((L-x)/\lambda)$	Reflection increases distal voltage

Conceptual Questions

Q11.1 The cable equation has three terms: capacitive current, axial diffusion, and membrane current. Explain the physical origin of each. Which term is absent in a point-neuron model?

Q11.2 Derive the cable equation by writing Kirchhoff's current law for a small segment Δx and taking the limit. What physical quantity does each term represent?

Q11.3 The length constant $\lambda = \sqrt{r_m a/(2r_L)}$ depends on the square root of the radius. What does this imply about the relative signal-carrying ability of thick versus thin dendrites? How much thicker must one dendrite be to have twice the length constant?

Q11.4 The membrane time constant $\tau_m = r_m c_m$ does not depend on the cable's geometry. Why not? What does this tell you about the relationship between spatial and temporal filtering in the cable?

Q11.5 For a steady-state infinite cable, the voltage decays as $e^{-\lvert x\rvert/\lambda}$. A synapse is located one length constant from the soma. What fraction of the synaptic voltage reaches the soma? What if the synapse is at 2λ? At 3λ?

Q11.6 A dendrite has $L/\lambda = 3$. Without computing exact numbers, describe qualitatively how a synaptic potential at the tip would appear at the soma compared to a synapse at the base.

Q11.7 Why does a sealed end increase the voltage at the tip compared to an infinite cable? What is the physical reason for this "reflection" of current?

Q11.8 The electrotonic length L/λ determines whether a cable behaves like a point neuron $(L/\lambda \ll 1)$ or shows significant spatial filtering $(L/\lambda \gg 1)$. What are the two ways a neuron could reduce its electrotonic length without changing its physical length?

Q11.9 Compare the cable equation for a passive membrane with the point-neuron membrane equation from Chapter 4. Under what conditions does the cable equation reduce to the point-neuron equation?

Q11.10 If the axial resistivity r_L is increased (for example, by reducing the concentration of mobile ions inside the cell), how does each cable parameter (λ, τ_m) change? What are the consequences for signal propagation?

Q11.11 A neuron has two dendrites: one thin $(a = 0.5\ \mu\text{m})$ and one thick $(a = 2\ \mu\text{m})$. Both receive identical synaptic inputs at the same physical distance from the soma. Which synapse has more influence on the somatic voltage? Why?

12

Cable Theory: Solutions and Branching

The passive cable equation has analytical solutions that reveal how signals are shaped by dendritic geometry. This chapter develops the time-dependent solution, analyzes branching structures, and introduces Rall's equivalent cylinder—a remarkable simplification that collapses an entire dendritic tree into a single cable.

12.1 Time-Dependent Solution

The steady-state solution from Section 11.4 tells us the final voltage distribution, but synaptic inputs are brief events. We need the full time-dependent response.

For an infinite passive cable with a brief current pulse (a delta function in both space and time) injected at $x = 0$ and $t = 0$, the voltage response is

$$v(x,t) = \frac{r_m}{2\lambda} \frac{1}{\sqrt{\pi t/\tau_m}} \exp\left(-\frac{t}{\tau_m} - \frac{x^2}{4\lambda^2 t/\tau_m}\right) \tag{12.1}$$

This is the Green's function of the passive cable. It has two factors: a *spatial Gaussian* whose width grows as $\sqrt{t/\tau_m}$, and an *exponential temporal decay* with time constant τ_m. The signal spreads out in space (diffusion) while simultaneously decaying in time (membrane leakage).

At the injection site ($x = 0$), the voltage rises instantaneously and then decays. At a distant point, the voltage rises after a delay, reaches a smaller peak, and then decays. The further from the injection site, the more delayed, attenuated, and broadened the response. This is the temporal filtering property of the cable: it acts as a low-pass filter, smoothing out fast transients.

12.1.1 Signal velocity in a passive cable

Signals in a passive cable do not propagate at a fixed velocity—they spread by diffusion. However, for practical purposes, one can define an effective propagation speed as the rate at which the peak of the voltage pulse advances. For large times, this is approximately

$$v_{\text{signal}} \approx \frac{2\lambda}{\tau_m} \tag{12.2}$$

For typical dendritic parameters ($\lambda \sim 0.5$ mm, $\tau_m \sim 20$ ms), this gives $v_{\text{signal}} \sim 0.05$ m/s—very slow compared to action potential propagation. This is why passive spread is adequate for short dendrites but not for long-range axonal signaling.

12.2 Asymmetric Attenuation

An important and initially counterintuitive feature of dendritic trees is that attenuation is *direction-dependent*. A signal traveling from a thin distal dendrite to the thick soma is attenuated much more than a signal traveling in the opposite direction.

The reason is impedance mismatch. The soma has a low input impedance (large membrane area, many attached dendrites). When current from a thin dendrite reaches the soma, it encounters a large conductance load that "sinks" the current, causing a large voltage drop. Conversely, the soma acts as a strong voltage source when driving current into a thin dendrite, and the high input impedance of the thin branch means even a small current produces a substantial voltage.

This asymmetry has functional consequences. Distal synaptic inputs are strongly filtered before reaching the soma, while back-propagating action potentials from the soma reach distal dendrites with relatively less attenuation. The neuron is not a symmetric filter—it treats inward and outward signals very differently.

12.3 Branching

Real dendritic trees branch extensively. At each branch point, two boundary conditions must hold: voltage is continuous across the junction, and current is conserved (Kirchhoff's law):

$$V_{\text{parent}} = V_{\text{daughter}_1} = V_{\text{daughter}_2}, \qquad I_{\text{parent}} = I_{\text{daughter}_1} + I_{\text{daughter}_2} \tag{12.3}$$

The axial current in each branch depends on its cross-sectional area: $I_a \propto a^2\, \partial V/\partial x$. The input impedance of a semi-infinite cable scales as $a^{-3/2}$, so the voltage distribution at a branch point depends on the relative radii of parent and daughter branches.

When current is injected into a thin daughter branch, it arrives at the branch point and encounters a low-impedance parent trunk (plus the other daughter). Most of the current is "absorbed" by the parent, and the voltage at the junction is small. The thin branch is effectively shunted by the thicker trunk.

The relative weight of each branch at a junction is proportional to $a_i^{3/2}$:

$$p_i = \frac{a_i^{3/2}}{\sum_j a_j^{3/2}} \tag{12.4}$$

12.3.1 Explicit three-branch solutions at an isolated junction

For a steady current I_e injected at position y along branch 2, the branch voltages can be written in closed form:

$$v_1(x_1) = p_1 I_e R_{\lambda,1} \exp\left(-\frac{x_1}{\lambda_1}\right) \exp\left(-\frac{y}{\lambda_2}\right), \tag{12.5}$$

$$v_3(x_3) = p_3 I_e R_{\lambda,3} \exp\left(-\frac{x_3}{\lambda_3}\right) \exp\left(-\frac{y}{\lambda_2}\right), \tag{12.6}$$

$$v_2(x_2) = \frac{I_e R_{\lambda,2}}{2} \left[\exp\left(-\frac{|x_2 - y|}{\lambda_2}\right) + (2p_2 - 1) \exp\left(-\frac{x_2 + y}{\lambda_2}\right) \right]. \tag{12.7}$$

The first term in Equation (12.7) is the infinite-cable contribution; the second term is the branch-point correction from impedance mismatch.

The coefficients p_i are fixed by continuity of voltage and conservation of axial current at the node:

$$\sum_i a_i^2 \left. \frac{\partial v_i}{\partial x_i} \right|_{x_i=0} = 0. \tag{12.8}$$

Using $R_{\lambda,i} \propto a_i^{-3/2}$, one obtains $p_i \propto 1/R_{\lambda,i}$ as expected from current partition into parallel impedances.

12.3.2 Impedance matching and differentiable continuation

A particularly important case is

$$p_1 = \frac{1}{2}, \qquad p_2 = p_3 = \frac{1}{4}, \tag{12.9}$$

which is equivalent to

$$2R_{\lambda,1} = R_{\lambda,2} = R_{\lambda,3}. \tag{12.10}$$

In this regime, the branch-point correction terms simplify and the transition through the node becomes differentiable in electrotonic coordinates. Geometrically, the branched cable behaves like a smooth continuation of a single effective cable—the local version of Rall's equivalent-cylinder idea.

Computationally, these explicit forms explain directional attenuation in quantitative terms: injecting current into a thin daughter branch produces stronger attenuation toward the node than injecting into the thick parent trunk.

12.4 Rall's Equivalent Cylinder

Wilfrid Rall showed that, under certain geometric constraints, an entire branching dendritic tree can be reduced to a single equivalent unbranched cylinder. This is a powerful simplification: it means the full passive behavior of a complex tree can be captured by a one-dimensional cable equation with appropriate parameters.

The key condition is the 3/2 *power law*: at every branch point, the sum of the daughter radii raised to the 3/2 power must equal the parent radius raised to the 3/2 power:

Rall's 3/2 Power Law

$$d_{\text{parent}}^{3/2} = \sum_k d_{\text{daughter}_k}^{3/2} \tag{12.11}$$

d: diameter of each branch. When this holds at every branch point, impedance is matched and no signal reflection occurs.

When the 3/2 law holds, the branching tree has the same input impedance as a single cylinder whose diameter matches the trunk and whose length is the electrotonic length of the tree. There is no reflection of signals at branch points—current flows smoothly through the tree as if it were unbranched.

The equivalent cylinder has the same total membrane area as the dendritic tree, and its electrotonic length L/λ is the sum of the electrotonic lengths of the segments from trunk to tip. The ends of the cylinder correspond to the sealed tips of the dendrites.

Real dendritic trees do not perfectly satisfy the 3/2 law, but many come close enough that the equivalent cylinder provides a useful approximation. Deviations from the law cause partial reflections at branch points, which can be analyzed as perturbations around the Rall solution.

Chapter Summary

Concept	Equation	Key Insight
Green's function	Gaussian × exponential decay	Signals spread (diffuse) and decay
Signal velocity	$v \approx 2\lambda/\tau_m$	Passive spread is slow
Asymmetric attenuation	Dendrite → soma: strong loss	Impedance mismatch is direction-dependent
Branch conditions	V continuous, I conserved	Kirchhoff's law at every junction
Three-branch solution	$v_i(x_i)$ from Equations (12.5) to (12.7)	Separates infinite-cable term and branch correction
Impedance matching	$2R_{\lambda,1} = R_{\lambda,2} = R_{\lambda,3}$	Node behaves like smooth cable continuation
Rall's 3/2 law	$d_{\text{parent}}^{3/2} = \sum d_k^{3/2}$	Impedance matching → equivalent cylinder

Conceptual Questions

Q12.1 The Green's function of the passive cable (Equation (12.1)) has both a Gaussian spatial factor and an exponential temporal decay. Explain the physical origin of each. What happens to the voltage pulse at a point far from the injection site?

Q12.2 A synaptic potential at a distal dendrite arrives at the soma with reduced amplitude and a delay. Using the Green's function, explain why the signal is not only smaller but also broader in time.

Q12.3 The passive cable acts as a low-pass filter: fast transients are smoothed more than slow ones. Explain why, using the relationship between the spatial spread of the Gaussian and the timescale of the signal.

Q12.4 Attenuation from dendrite to soma is stronger than from soma to dendrite. Why? Use the concept of impedance mismatch in your answer.

Q12.5 A back-propagating action potential from the soma reaches distal dendrites with relatively little attenuation compared to the reverse direction. What functional role might this asymmetry play in synaptic plasticity?

Q12.6 At a branch point, voltage is continuous and current is conserved. Why do these two conditions fully determine the voltage distribution? What would happen if current were not conserved—what physical situation would that represent?

Q12.7 The weight of each branch at a junction is proportional to $a_i^{3/2}$. Why 3/2 specifically, and not a^2 (the cross-sectional area)? What role does the input impedance play?

Q12.8 Rall's 3/2 power law says $d_{\text{parent}}^{3/2} = \sum d_k^{3/2}$ at every branch point. What goes wrong if this condition is violated? What happens to a signal arriving at a branch point that does not satisfy impedance matching?

Q12.9 The equivalent cylinder model collapses a complex tree into a single cable. What information about the original tree is preserved, and what is lost? Can two different dendritic trees have the same equivalent cylinder?

Q12.10 A neuron has a long, thin dendrite that branches into two equally thin daughters. Does signal attenuation increase or decrease at the branch point compared to an unbranched continuation? Why?

Q12.11 The signal velocity in a passive cable is $v_{\text{signal}} \approx 2\lambda/\tau_m$. For typical dendritic parameters, this is about 0.05 m/s. Why is this far too slow for long-range axonal communication, and how does this motivate the need for active (action potential) propagation?

Q12.12 Compare the passive cable equation with the one-dimensional diffusion equation. What is the analogy between voltage spread in a cable and heat diffusion in a rod? Where does the analogy break down?

13

Compartment Models, Propagation, and Myelination

The analytical solutions of cable theory apply only to passive, uniform cables. Real neurons have complex morphologies and nonlinear channels. This chapter introduces the compartmental method for numerical simulation, analyzes action potential propagation along axons, and shows how myelination dramatically increases conduction velocity.

13.1 Multi-Compartment Models

The continuous cable equation is a partial differential equation that admits analytical solutions only in special cases (passive membrane, uniform geometry). To handle realistic neurons—with nonlinear channels, varying diameters, and complex branching—we discretize the cable into a chain of compartments.

Each compartment n is a short cylinder of length Δx, assumed isopotential (uniform voltage within the compartment). Adjacent compartments are coupled by axial conductances. The voltage in compartment n evolves according to

$$C_m \frac{\mathrm{d}V_n}{\mathrm{d}t} = -I_{\text{ion}}(V_n, t) + \sum_k G_{n,k}(V_k - V_n) + I_{\text{inj}} \tag{13.1}$$

where $I_{\text{ion}}(V_n, t)$ is the total membrane current (which can be nonlinear—Hodgkin–Huxley channels, synaptic currents, etc.), $G_{n,k}$ is the axial conductance between compartments n and k, and I_{inj} is any injected current.

The compartmental method converts the PDE into a system of coupled ODEs—one per compartment—that can be integrated numerically. The spatial resolution must be fine enough to capture the relevant voltage gradients. A standard rule of thumb is $\Delta x \lesssim \lambda/5$; coarser discretization introduces spatial artifacts. For a typical neuron with dendrites a few hundred micrometers long and $\lambda \sim 0.5$ mm, this means compartments of roughly 100 μm or less.

The advantage of compartmental models is generality. Any channel type can be placed in any compartment, the geometry can be reconstructed from anatomical data, and the simulation is straightforward. The cost is that you lose the analytical insight of continuous cable theory.

13.2 Action Potential Propagation

In the passive cable, signals spread by diffusion—they decay exponentially and have no fixed velocity. Action potential propagation is qualitatively different: the spike travels as a *wave* that regenerates itself at each point along the axon, maintaining constant amplitude and shape.

The mechanism is local positive feedback. At the wavefront, the voltage is rising, which opens sodium channels in the adjacent (resting) membrane. The inward sodium current depolarizes that region, opening more channels, and the wavefront advances. Behind the wavefront, the membrane is in its refractory period (sodium inactivated, potassium still open), so the wave cannot travel backward.

13.2.1 Propagation speed in unmyelinated axons

The speed of the action potential depends on how quickly the wavefront can depolarize the next stretch of membrane. The relevant timescale is τ_m (how fast the membrane charges) and the relevant length scale is λ (how far ahead the axial current reaches). Dimensional analysis gives

AP Propagation Speed (Unmyelinated)

$$v_{\mathrm{AP}} \approx \frac{2\lambda}{\tau_m} \propto \sqrt{a} \tag{13.2}$$

$\lambda \propto \sqrt{a}$: length constant, $\tau_m = r_m c_m$: membrane time constant (independent of a).

Speed scales as $\sqrt{a}$: doubling the speed requires quadrupling the radius.

For typical unmyelinated axon parameters, this gives $v_{\mathrm{AP}} \sim 0.1$–1 m/s. The squid giant axon achieves high speed (~ 25 m/s) by having an enormous radius (~ 0.5 mm). This brute-force strategy is metabolically expensive and takes up a lot of space. Vertebrates use a different solution: myelination.

13.2.2 Wave properties

Action potential waves have several notable properties. They travel in both directions from the initiation site (orthodromic and antidromic propagation). Two colliding waves annihilate each other, because the membrane behind each wavefront is refractory. Waves do not reflect from sealed ends or branch points (though they can fail to propagate past a branch point if

the impedance load of the daughters is too large).

13.3 Myelination

Myelin is a lipid-rich insulating sheath wrapped around axons by glial cells (oligodendrocytes in the central nervous system, Schwann cells in the peripheral nervous system). The myelin is interrupted at regular intervals by *nodes of Ranvier*, where the axon membrane is exposed and densely packed with sodium channels.

Myelination changes two membrane parameters. The insulating wraps increase the effective membrane resistance r_m^* (less leakage through the sheath) and decrease the effective membrane capacitance c_m^* (the myelin acts as a series capacitance). For a sheath with inner radius a_1 (the axon) and outer radius a_2 (including myelin):

$$r_m^* = r_m \frac{a_1}{d} \ln\left(\frac{a_2}{a_1}\right), \qquad c_m^* = c_m \frac{d}{a_1 \ln(a_2/a_1)} \tag{13.3}$$

where d is the membrane thickness. The product $r_m^* c_m^* \approx r_m c_m$, so the time constant is approximately unchanged. But the length constant increases dramatically: $\lambda^* \propto \sqrt{r_m^*} \gg \lambda$.

13.3.1 Reduced myelinated-cable scaling

For an internode segment (no active regeneration), the cable equation becomes

$$r_m^* c_m^* \frac{\partial v}{\partial t} = \frac{r_m^* a_1}{2 r_L} \frac{\partial^2 v}{\partial x^2} - v. \tag{13.4}$$

On internodal timescales where longitudinal spread dominates local leak, the $-v$ term is subleading, giving an effective diffusion equation:

$$\frac{\partial v}{\partial t} = D_* \frac{\partial^2 v}{\partial x^2}, \qquad D_* = \frac{a_1}{2 r_L c_m^*} = \frac{a_1^2 \ln(a_2/a_1)}{2 r_L c_m d}. \tag{13.5}$$

Thus, geometry controls the dominant propagation scale through the factor $a_1^2 \ln(a_2/a_1)$.

At fixed outer radius a_2, maximizing this factor yields $a_1/a_2 = 1/\sqrt{e}$, exactly the optimum in Equation (13.7). This provides a compact derivation of why the same ratio emerges from conduction-speed and electrotonic-length arguments.

13.3.2 Saltatory conduction

Conduction in myelinated axons is *saltatory*: the action potential "jumps" from node to node. At each node, voltage-gated sodium channels regenerate the spike. Between nodes, current

spreads passively through the myelinated internode with minimal leakage and minimal capacitive loading. The result is much faster conduction.

The key change is that the propagation speed now scales linearly with the outer radius rather than as its square root:

$$v_{\mathrm{AP}}^{\mathrm{myel}} \propto a_2 \tag{13.6}$$

This is a dramatic improvement. A myelinated axon of 10 μm diameter can conduct at 50–100 m/s, comparable to the squid giant axon at 500 μm diameter. Myelination achieves the same speed with $\sim$ 2500 times less cross-sectional area.

13.3.3 Empirical anchoring

Peripheral-nerve measurements (e.g., cat fibers) show an approximately linear conduction-velocity trend with fiber diameter over a broad range, consistent with the scaling in Equation (13.6). A practical rule of thumb is that each additional micrometer of myelinated fiber diameter contributes several meters per second of conduction speed.

The linear trend is not exact—node spacing, temperature, and channel density also matter—but the first-order dependence on outer diameter is robust and is one of the clearest biophysical signatures of saltatory conduction.

Computationally, this empirical relation explains why long-range pathways can simultaneously be fast and compact: conduction delay can be tuned by modest anatomical changes without requiring giant axons.

13.3.4 Optimal myelin thickness

How thick should the myelin be? Thicker myelin increases r_m^* and λ^*, but it also increases the outer radius a_2, which takes up space. There is an optimum ratio of inner to outer radius that maximizes the length constant for a fixed outer diameter:

$$\frac{a_1}{a_2} = \frac{1}{\sqrt{e}} \approx 0.6 \tag{13.7}$$

This means the axon should occupy about 60% of the total fiber diameter, with 40% being myelin. Remarkably, measurements of real myelinated axons consistently find ratios of 0.6–0.7, close to the theoretical optimum. Biology has found the solution predicted by the mathematics.

Chapter Summary

Concept	Equation	Key Insight
Compartmental model	$C_m \dot{V}_n = -I_{\text{ion}} + \sum G_{n,k}(V_k - V_n)$	Discretize cable for numerical simulation
AP speed (unmyelinated)	$v \approx 2\lambda/\tau_m \propto \sqrt{a}$	Square-root scaling: slow without myelin
Myelin effect	$r_m^* \uparrow, c_m^* \downarrow, \lambda^* \uparrow$	Less leakage, less capacitance
Reduced internode dynamics	$\partial_t v = D_* \partial_x^2 v$	Geometry-controlled rapid passive spread between nodes
Saltatory conduction	Node-to-node regeneration	Speed $\propto a_2$ (linear scaling)
Optimal ratio	$a_1/a_2 = 1/\sqrt{e} \approx 0.6$	Biology matches the theoretical optimum

Conceptual Questions

Q13.1 The compartmental model converts the cable PDE into a system of coupled ODEs. What is lost in this discretization? What determines how fine the spatial grid must be?

Q13.2 In a compartmental model, what determines how fine the spatial discretization must be? What goes wrong if Δx is too large compared to λ?

Q13.3 Compare the compartmental model to the equivalent cylinder approach from Chapter 12. What are the advantages and limitations of each?

Q13.4 Action potential propagation is fundamentally different from passive spread: the spike maintains its amplitude as it travels. Explain the mechanism of regeneration at each point along the axon.

Q13.5 Why does the action potential travel in only one direction under normal physiological conditions, even though the physics would allow bidirectional propagation? What prevents backward propagation?

Q13.6 Action potential speed in unmyelinated axons scales as $\sqrt{a}$, but in myelinated axons it scales as a. Explain the physical reason for this difference.

Q13.7 The squid giant axon achieves fast conduction by having a very large radius (~ 0.5 mm). A myelinated vertebrate axon of only 10 μm diameter conducts at comparable speed. Calculate roughly how much cross-sectional area the myelinated solution saves.

Q13.8 Myelination increases r_m^* and decreases c_m^*, but the product $r_m^* c_m^*$ stays roughly constant. Why does the conduction speed increase if the time constant does not change?

Q13.9 In saltatory conduction, the action potential "jumps" from node to node. What happens at each node? What happens in the internode? Why is this faster than continuous regeneration?

Q13.10 The optimal ratio $a_1/a_2 \approx 0.6$ maximizes the length constant for a fixed outer diameter. What happens if the myelin is too thin (ratio near 1)? What if it is too thick (ratio near 0)?

Q13.11 The optimal ratio $a_1/a_2 = 1/\sqrt{e}$ is derived from maximizing λ^* at fixed outer radius. Measurements of real axons give ratios of 0.6–0.7. What does this agreement between theory and biology suggest about evolutionary optimization?

Q13.12 Two action potential waves traveling toward each other on the same axon annihilate when they meet. Why? What is the state of the membrane at the collision point?

Q13.13 Demyelinating diseases like multiple sclerosis strip myelin from axons. Using what you know about the effects of myelination on r_m^*, c_m^*, and λ^*, predict the consequences for conduction speed. At what point would conduction fail entirely?

14

Recurrent Network Models

Single neurons compute, but brains are networks. This chapter develops the firing-rate model of neural populations, analyzes linear recurrent networks using eigenmode decomposition, and shows how recurrent connectivity can amplify, select, and sustain patterns of activity. The ring network for head-direction coding illustrates these principles in a concrete biological setting.

14.1 From Spikes to Firing Rates

Everything in this course so far has described individual neurons: their membrane potentials, channel kinetics, and cable properties. But the brain contains billions of neurons operating together. To study networks, we need to shift from tracking individual spikes to tracking *population activity*—the firing rate of groups of neurons.

The simplest version of this transition replaces the detailed spike-generating dynamics with a static input–output function. If the synaptic input current to a neuron varies slowly compared to the membrane time constant, the neuron's firing rate tracks the steady-state *f–I* curve:

$$v(t) = F(I_s(t)) \tag{14.1}$$

where F is the activation function—the neuron's *f–I* curve, which maps input current to firing rate. For Type I neurons, F rises smoothly from zero; for Type II neurons, it jumps to a nonzero value.

This steady-state approximation works well when inputs change slowly. For fast-varying inputs, the neuron cannot keep up instantaneously: there is a lag between changes in input and changes in firing rate. The Wilson–Cowan model accounts for this by adding a phenomenological time constant:

$$\tau_r \frac{\mathrm{d}v}{\mathrm{d}t} = -v + F(I(t)) \tag{14.2}$$

The time constant τ_r is not simply the membrane time constant τ_m—it depends on input strength, noise, and network context. It represents the lag between changes in input and changes in population firing rate.

14.2 Network Architecture

A network of N firing-rate neurons has a state vector $\mathbf{v}(t) \in \mathbb{R}^N$ (the firing rates of all neurons) driven by an input vector $\mathbf{u}(t)$ (external drive). The network is defined by two weight matrices: a *feedforward* matrix W that maps inputs to internal drive, and a *recurrent* matrix M that describes how neurons influence each other.

General Recurrent Rate Model

$$\tau_r \dot{\mathbf{v}} = -\mathbf{v} + F(W\mathbf{u} + M\mathbf{v}) \tag{14.3}$$

$\mathbf{v}$: firing rate vector, $\mathbf{u}$: external input, W: feedforward weights, M: recurrent weights, F: activation function.

The feedforward term $W\mathbf{u}$ transforms and filters the input. The recurrent term $M\mathbf{v}$ feeds the network's own activity back into itself. The interplay between feedforward drive and recurrent feedback determines the network's computational behavior.

14.3 Linear Recurrent Networks

The simplest case sets F to the identity function, giving a linear system:

$$\tau_r \dot{\mathbf{v}} = -\mathbf{v} + \mathbf{h} + M\mathbf{v} \tag{14.4}$$

where $\mathbf{h} = W\mathbf{u}$ is the feedforward drive. Despite its simplicity, this linear model reveals the essential logic of recurrent computation.

14.3.1 Eigenmode decomposition

If M is symmetric (a simplification; real connectivity is not), its eigenvalues λ_μ are real and its eigenvectors $\mathbf{e}_\mu$ are orthogonal. Expanding the state as $\mathbf{v}(t) = \sum_\mu c_\mu(t)\,\mathbf{e}_\mu$, each mode decouples:

Eigenmode Dynamics

$$\tau_r \dot{c}_\mu = -(1 - \lambda_\mu)\, c_\mu + \mathbf{e}_\mu \cdot \mathbf{h} \tag{14.5}$$

$c_\mu(t)$: amplitude of mode μ, λ_μ: eigenvalue of M, $\mathbf{e}_\mu \cdot \mathbf{h}$: projection of input onto mode μ.

Each mode evolves independently as a simple first-order equation. The effective time constant is $\tau_r/(1-\lambda_\mu)$, and the steady-state amplitude (for stable modes) is $c_\mu(\infty) = (\mathbf{e}_\mu \cdot \mathbf{h})/(1-\lambda_\mu)$.

14.3.2 Stability, amplification, and integration

The eigenvalue λ_μ controls everything about mode μ:

If $\lambda_\mu < 1$, the mode is stable. The input component along $\mathbf{e}_\mu$ is amplified by a gain factor $1/(1-\lambda_\mu)$, and the effective time constant is stretched by the same factor. Eigenvalues close to 1 produce strong amplification and slow dynamics.

If $\lambda_\mu > 1$, the mode is unstable: the corresponding pattern of activity grows exponentially. In a real network, nonlinearities (saturation, rectification) eventually bound the growth, but in the linear model, this represents an explosion.

If $\lambda_\mu = 1$ exactly, the leak and recurrent feedback cancel perfectly, and the mode becomes a *perfect integrator*:

$$c_1(t) = \frac{1}{\tau_r} \int_0^t \mathbf{e}_1 \cdot \mathbf{h}(t')\ \mathrm{d}t' \tag{14.6}$$

The network accumulates input along $\mathbf{e}_1$ without decay—a form of persistent activity or working memory. But this requires $\lambda_1 = 1$ with infinite precision. If $\lambda_1 = 0.99$, the memory leaks away; if $\lambda_1 = 1.01$, activity explodes. This *fine-tuning problem* is a central challenge for linear models of persistent activity.

14.3.3 Selective amplification

When one eigenvalue is close to 1 and the others are small, the network strongly amplifies the input component that matches the corresponding eigenvector, while other components are only weakly amplified or suppressed. The network acts as a "matched filter"—it selectively extracts the pattern it is tuned to represent.

14.4 The Ring Network

The ring network is a concrete example that ties together selective amplification, pattern formation, and biological function. It models neurons arranged in a circle, each with a preferred direction (as in the head-direction system of the fly or the mammalian thalamus).

The recurrent connectivity depends on the angular difference between neurons:

$$M(\theta, \theta') \propto \cos(\theta - \theta') \tag{14.7}$$

Nearby neurons (similar preferred directions) excite each other; distant neurons inhibit each other. This "Mexican hat" connectivity profile has a clean eigenstructure: the eigenvectors are Fourier modes (cosines and sines), and the fundamental mode (a single bump) has the largest eigenvalue.

If the fundamental eigenvalue is set near 1 while higher harmonics have small eigenvalues, the network selectively amplifies broad, smooth patterns of activity. A noisy input with multiple peaks is cleaned up: the network sustains a single bump of activity centered on the most strongly driven direction. This bump represents the animal's current heading.

14.4.1 Persistent activity and path integration

In darkness, with no sensory input, the bump persists—a memory of the last known heading. This is the ring attractor functioning as a working memory for a continuous variable. The bump can be shifted by vestibular or motor signals (velocity inputs), implementing *path integration*: updating the heading estimate based on self-motion.

Experimental recordings from the fly's ellipsoid body confirm this picture: calcium imaging reveals a bump of activity that rotates with the fly's heading, persists in darkness, and drifts slowly without visual landmarks (confirming that the system is an imperfect integrator).

Comparable head-direction populations in rodents show related behavior: internally generated heading signals are stable over short intervals but accumulate drift without external reference cues. Visual and landmark input then acts as an error-correction channel, pulling the bump back toward veridical heading.

This drift-correction interplay is computationally central. Pure path integration accumulates error; pure sensory tracking fails when cues are absent. The ring network provides a principled fusion architecture where recurrent dynamics maintain memory and sensory input periodically re-anchors it.

14.5 Nonlinear Networks: Rectification and Winner-Take-All

Linear networks have a serious problem: firing rates can be negative, which is unphysical. Adding a rectifying nonlinearity—$F(x) = [x]_+ = \max(0, x)$—fixes this and fundamentally changes the network's behavior.

Rectification turns amplification into *selection*. In a linear network, all modes are amplified (some more than others). With rectification, the strongly amplified mode drives competing neurons' rates to zero, and only the winning pattern survives. This is winner-take-all dynamics: the network selects the single input direction that best matches its recurrent structure and suppresses everything else.

In the ring network, rectification sharpens the activity bump and creates a robust attractor. The bump can sit at any angle on the ring—the set of all possible bump positions forms a *ring attractor*. Unlike the fine-tuned linear integrator, the nonlinear ring attractor sustains activity without requiring $\lambda = 1$ exactly. Nonlinearity provides the robustness that linear models lack.

The feedforward weights W determine the initial response to a sensory input (its "shape"), while the recurrent weights M govern what happens afterward: amplification, selection, and sustained memory through feedback.

14.5.1 Beyond winner-take-all: gain modulation, associative memory, and oscillations

Rectified recurrent networks support more than input selection. Three extensions are especially useful:

1. **Gain modulation**: recurrent context can scale response amplitude without changing preferred feature, implementing multiplicative-like computations used in sensorimotor transformations.

2. **Associative memory**: with structured recurrent weights (e.g., Hebbian imprinting of patterns), activity can converge to stored attractors, enabling pattern completion from partial cues.

3. **Oscillatory regimes**: with suitable excitation–inhibition balance and delays, recurrent networks can generate rhythmic population activity rather than fixed-point attractors.

These behaviors are not separate model classes; they are different operating regimes of the same recurrent architecture under different nonlinearities and connectivity structure.

Computationally, this is why recurrent models are versatile: the same circuit motif can denoise, select, integrate, memorize, and synchronize depending on parameter regime and

input context.

Chapter Summary

Concept	Equation	Key Insight
Rate model	$\tau_r \dot{\mathbf{v}} = -\mathbf{v} + F(W\mathbf{u} + M\mathbf{v})$	Feedforward drive + recurrent feedback
Eigenmode	$\tau_r \dot{c}_\mu = -(1-\lambda_\mu)c_\mu + \mathbf{e}_\mu \cdot \mathbf{h}$	Each mode evolves independently
Amplification	$\text{Gain} = \dfrac{1}{1-\lambda_\mu}$	$\lambda \to 1$: strong amplification, slow dynamics
Perfect integrator	$\lambda = 1$: no decay	Working memory, but requires fine-tuning
Ring connectivity	$M \propto \cos(\theta - \theta')$	Head-direction bump, ring attractor
Rectification	$F(x) = \max(0, x)$	Selection, winner-take-all, robust attractors
Drift-corrected path integration	Internal integration + landmark anchoring	Persistent memory with external error correction
Nonlinear recurrent extensions	Gain modulation / attractor memory / oscillations	One architecture, multiple computational regimes

Conceptual Questions

Q14.1 The transition from spikes to rates involves replacing the detailed spike dynamics with a static *f–I* curve. Under what conditions is this approximation valid? When does it break down?

Q14.2 The Wilson–Cowan equation $\tau_r \dot{v} = -v + F(I)$ introduces a time constant τ_r that is not

simply the membrane time constant. What determines τ_r, and why is it generally faster than τ_m?

Q14.3 In a linear recurrent network, why does an eigenvalue close to 1 produce both strong amplification and slow dynamics? What is the connection between these two effects?

Q14.4 For a linear network with eigenvalues $\lambda_1 = 0.95, \lambda_2 = 0.5, \lambda_3 = 0.1$, compute the gain and effective time constant for each mode. Which mode dominates the response?

Q14.5 The "fine-tuning problem" says that $\lambda = 1$ is required for perfect integration but is biologically unrealistic. Why is exact fine-tuning unrealistic? What happens to the "memory" if $\lambda = 0.99$ versus $\lambda = 1.01$?

Q14.6 A linear recurrent network with one eigenvalue near 1 and all others small acts as a "matched filter." What does it filter for, and how does the eigenvector structure determine this selectivity?

Q14.7 Explain how the ring network implements a head-direction system. What does the "bump" of activity represent, and how is it maintained and updated?

Q14.8 The ring network has connectivity $M(\theta, \theta') \propto \cos(\theta - \theta')$. Why do the eigenvectors of this connectivity matrix correspond to Fourier modes? Which mode has the largest eigenvalue?

Q14.9 In darkness, the head-direction bump persists but drifts slowly. What does this drift tell you about how close the network's eigenvalue is to 1? How could the brain correct for this drift?

Q14.10 How does nonlinearity (rectification) solve the fine-tuning problem in the ring network? Why does a rectified bump attractor not require $\lambda = 1$ exactly?

Q14.11 Compare the roles of the feedforward matrix W and the recurrent matrix M. Which determines the initial response to a stimulus, and which determines the long-term dynamics? Give an example where each dominates.

Q14.12 Winner-take-all dynamics emerge from the combination of recurrent excitation and rectification. Explain the mechanism: why does the strongest input suppress all others?

Q14.13 A ring attractor can store a continuous variable (like heading angle). How is this different from a discrete attractor network that stores one of a finite number of patterns? What biological computations might require each type?

Q14.14 Why is path integration alone insufficient for long timescales? How do sensory landmarks correct the drift of a recurrent bump representation?

Q14.15 Give one mechanism by which the same recurrent network could move from winner-take-all behavior to oscillatory behavior.

Q14.16 The eigenmode decomposition assumes M is symmetric. Real neural connectivity is typically asymmetric ($M_{ij} \neq M_{ji}$). What qualitatively new phenomena can asymmetric connectivity produce that symmetric connectivity cannot?

Part V

Stochastic Neural Dynamics

15

Stochastic Spiking and Neural Noise

Deterministic membrane equations predict a unique trajectory for each input, but real neurons fire variably even under repeated stimulation. This chapter introduces the statistical language needed to analyze stochastic spike trains: interspike-interval distributions, count variability, spectral measures, and signal-transmission metrics such as coherence and information rate. The goal is to connect single-neuron noise statistics to computational function.

15.1 Why a Stochastic Description?

In Chapters 5, 6 and 10, spikes emerged from deterministic dynamics. If we repeat the same current pulse in those models, we get the same spike train each trial. Real neurons are different: repeated trials show jitter in spike timing, variable spike counts, and spontaneous firing without explicit stimulus.

The sources of variability are multiple: channel noise from stochastic gating (Chapter 7), synaptic release noise, fluctuating network input, and hidden state variables not included in reduced models. A deterministic equation remains useful for the mean trajectory, but statistical descriptors are needed for variability around that mean.

Computationally, this shift matters because information is carried not only by mean firing rates but also by timing precision and trial-to-trial variability. Two neurons with identical mean rates can have very different coding performance if their spike statistics differ.

15.2 Spike Trains as Point Processes

A spike train is represented as a sum of Dirac impulses at spike times $\{t_k\}$:

$$x(t) = \sum_k \delta(t - t_k) \tag{15.1}$$

This turns event times into a signal suitable for averaging, spectrum analysis, and linear-response theory.

Define the interspike intervals (ISIs) $T_k = t_{k+1} - t_k$. Their distribution $p_{\mathrm{ISI}}(T)$ is the first

statistical object. The mean firing rate is

$$r = \frac{1}{\langle T \rangle}. \tag{15.2}$$

The coefficient of variation of ISIs is

$$\mathrm{CV} = \frac{\sqrt{\mathrm{Var}[T]}}{\langle T \rangle}. \tag{15.3}$$

For near-periodic firing, CV $\ll 1$; for irregular firing, CV ~ 1; for bursty or heavy-tailed statistics, CV > 1.

Computationally, ISI statistics separate distinct coding regimes: reliable temporal coding requires low-jitter ISIs, while high-variability regimes can support probabilistic sampling or exploration.

15.3 Count Variability and the Fano Factor

Let N_T be the number of spikes in a counting window of duration T. The Fano factor is

Fano Factor

$$F(T) = \frac{\mathrm{Var}[N_T]}{\mathbb{E}[N_T]}. \tag{15.4}$$

$F = 1$: Poisson-like variability, $F < 1$: more regular than Poisson, $F > 1$: overdispersed (e.g., bursts, slow-rate fluctuations).

For renewal processes, short and long windows can yield different behavior; slow gain fluctuations typically inflate $F(T)$ at large T. This scale dependence is important experimentally: a neuron may look near-Poisson at 20 ms and strongly overdispersed at 1 s.

Computationally, the Fano factor measures reliability of rate codes. Lower F improves detectability of weak stimulus-driven rate changes in finite observation windows.

15.4 Spectra, Coherence, and Signal Transmission

When spikes encode time-varying signals, second-order frequency-domain statistics become central. Let $s(t)$ be a stimulus and $x(t)$ the spike train. Define auto- and cross-spectra $S_{xx}(f)$, $S_{ss}(f)$, and $S_{xs}(f)$.

The coherence between stimulus and spike train is

$$C_{xs}(f) = \frac{|S_{xs}(f)|^2}{S_{xx}(f)S_{ss}(f)}, \qquad 0 \le C_{xs}(f) \le 1. \tag{15.5}$$

$C_{xs}(f)$ quantifies how well spike fluctuations at frequency f are locked to stimulus fluctuations at the same frequency. The phase of $S_{xs}(f)$ gives the frequency-dependent lag.

A commonly used lower bound on mutual-information rate for Gaussian channels is

$$I \ge -\int_0^\infty \log_2(1 - C_{xs}(f))\, \mathrm{d}f. \tag{15.6}$$

Computationally, this framework makes temporal coding measurable: resonator neurons often show coherence peaks near preferred frequencies, while integrator-like neurons transmit more broadband low-frequency content.

15.5 Noise-Driven Spiking in Reduced Models

A minimal stochastic extension of the LIF model adds a fluctuating input term:

$$\tau_m \frac{\mathrm{d}V}{\mathrm{d}t} = -(V - E_\mathrm{L}) + R_m[I_0 + \sigma\xi(t)], \tag{15.7}$$

with threshold/reset as usual and white noise $\xi(t)$ satisfying $\langle\xi(t)\rangle = 0$ and $\langle\xi(t)\xi(t')\rangle = \delta(t - t')$.

Even when I_0 is subthreshold, noise can trigger spikes (escape over an effective barrier). The result is a nonzero spontaneous firing rate with broad ISI statistics. As σ increases, CV and Fano factor typically increase, and timing precision relative to fast signals decreases.

Computationally, this connects dynamical excitability to variability statistics: bifurcation type sets deterministic backbone, while noise sets operating-point reliability around that backbone.

15.6 Spontaneous vs Evoked Activity

In spontaneous activity, statistics are characterized without controlled input: baseline rate, ISI density, CV, Fano factor, and spike-train spectrum. In evoked activity, we additionally measure transfer properties: gain, phase lag, coherence, and information rate.

A practical workflow is: (1) estimate baseline renewal statistics, (2) apply weak sinusoidal or stochastic stimuli, (3) compute linear response and coherence, (4) compare observed coding limits to predictions from the baseline noise model.

This workflow links what looks like "noise" to what is actually an important determinant of neural computation. Noise sets detection thresholds, response latency distributions, and the precision with which time-varying stimuli can be represented.

Chapter Summary

Concept	Equation	Key Insight
Spike train representation	$x(t) = \sum_k \delta(t - t_k)$	Event times as analyzable signal
Rate from ISIs	$r = 1/\langle T \rangle$	Mean interval sets mean firing rate
ISI irregularity	$\mathrm{CV} = \sqrt{\mathrm{Var}[T]}/\langle T \rangle$	Dimensionless timing variability
Count variability	$F(T) = \mathrm{Var}[N_T]/\mathbb{E}[N_T]$	Reliability of rate code in finite windows
Coherence	$C_{xs}(f) = \lvert S_{xs}\rvert^2/(S_{xx}S_{ss})$	Frequency-specific stimulus locking
Stochastic LIF	$\tau_m \dot{V} = -(V - E_{\mathrm{L}}) + R_m(I_0 + \sigma\xi)$	Noise converts deterministic threshold to probabilistic spiking

Conceptual Questions

Q15.1 Two neurons have the same mean firing rate but different CV values (0.2 vs 1.1). What distinct coding regimes do these values suggest?

Q15.2 Why can the Fano factor depend strongly on counting-window duration T? What mechanism produces large-T overdispersion?

Q15.3 Coherence $C_{xs}(f)$ is high near 10 Hz and low elsewhere. What does this say about the neuron's temporal selectivity and likely excitability class?

Q15.4 In the stochastic LIF model, how can a neuron fire when I_0 is subthreshold? How does increasing σ qualitatively affect ISI statistics?

Q15.5 Why is it useful to analyze both spontaneous and evoked activity rather than only one of them?

Bibliography

[DA01] Peter Dayan and Laurence F. Abbott. *Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems*. MIT Press, 2001.

[Ger+14] Wulfram Gerstner, Werner M. Kistler, Richard Naud, and Liam Paninski. *Neuronal Dynamics: From Single Neurons to Networks and Models of Cognition*. Cambridge University Press, 2014.

[HH52] Andrew L. Hodgkin and Andrew F. Huxley. "A quantitative description of membrane current and its application to conduction and excitation in nerve". In: *The Journal of Physiology* 117.4 (1952), pp. 500–544.

[Izh07] Eugene M. Izhikevich. *Dynamical Systems in Neuroscience: The Geometry of Excitability and Bursting*. MIT Press, 2007.

[Koc99] Christof Koch. *Biophysics of Computation: Information Processing in Single Neurons*. Oxford University Press, 1999.

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